

Exact Closure Alternatives for a Fixed-Shift Prime Packet: Quantifier-Safe Gates, Shorted-Gram Dominance, and Localized Alias Obstructions

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Abstract

The fixed-shift TPC packet now has exact coordinate, selection, prime-fiber, native-alias, high-face, missing-reassembly, and cofactor-exposure interfaces, but their conclusions carry different quantifiers and several are only certificates for one proof architecture. We give a quantifier-safe closure calculus for this chain.

For a nonzero positive semidefinite parent form R and a final physical map F , let $N = \text{Ker } R$, $S = N^\perp$, and $W = F(N)$. The optimal coefficient-uniform transfer constant is exactly

$$\alpha(F; R) = \lambda_{\min} \left(R_S^{-1/2} F_S^* P_{W^\perp} F_S R_S^{-1/2} \right),$$

and $\alpha(F; R) > 0$ if and only if $\text{Ker } F \subseteq \text{Ker } R$. Using classical shorted-operator theory, with self-contained finite-dimensional proofs, we apply transitivity on nested retained spaces: sequential elimination of genuine nuisance directions equals one master shorting, whereas ordinary compression can be strictly wrong. This separates mandatory kernel gates from sufficient certificates whose failure proves nothing about the final map.

For a finite selector family (S_a) , we prove the strict quantifier gap

$$\sup_a \inf_{z \neq 0} \frac{\|S_a z\|^2}{\|z\|^2} \leq \inf_{z \neq 0} \sup_a \frac{\|S_a z\|^2}{\|z\|^2}.$$

Thus an adaptive good phase sector or dyadic block need not produce one frozen selector with a uniform lower bound. Retaining the selector label by direct sum gives the exact uniform constant $\lambda_{\min}(\sum_a S_a^* S_a)$, but erasing that label is a new reassembly problem.

We also order the reassembly certificates. The master shorted constant dominates every reverse-triangle lower certificate; the physical restricted Gram dominates both residual-factor and ambient-degree relaxations, while those two coarse relaxations are generally incomparable. Finally, if the unit-column missing Gram has $\kappa_M \geq 1 + L$, then some row and one of B literal correlation cells carry alias mass at least L/B . With at most Δ neighbors in that cell, one normalized pair has magnitude at least $L/(B\Delta)$. Hence overlap degree $X^{\delta+o(1)}$ and pair correlation $X^{-\tau+o(1)}$ imply $\nu_M \leq (\delta - \tau)_+$. On the TPC-61 carrier these entries are the explicit fixed- h_0 two-affine Möbius/determinant sums. This localizes a large alias obstruction; it does not estimate a Möbius correlation.

A three-gate theorem closes only the declared truncated packet when the represented-chain, reassembly, and downstream losses are proved in the same scope and sum to less than $1/400$. We give sharp independence models and kill/pivot rules. No fixed-shift parity improvement, prime-pair lower bound, or twin-prime consequence is asserted.

Keywords: fixed shift; shorted Gram; quantifier order; adaptive selector; certificate dominance; alias localization.

Convention. All Hilbert spaces are finite dimensional at each scale. Every physical specialization uses one fixed nonzero shift h_0 , literal coefficients, and raw counting-measure atomic norms. The symbol h_0 is not specialized to 2; such a specialization and its number-theoretic target identification are not results of this paper.

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1 Introduction

The papers TPC–53–TPC–61 replace a long heuristic transfer by explicit finite-dimensional objects on one fixed nonzero physical shift [3–11]. They identify literal atom coefficients, sharp selection constants, prime-resolved singular values, adaptive phase sectors, undeleted singleton floors, residual and native alias Grams, a critical very-high cutoff, missing-coordinate reassembly, realized cofactor exposure, and finally a normalized two-affine Möbius kernel. Most of these objects are exact. That does not mean they form one proved lower-bound chain.

There are three reasons.

1. Some results are coefficient-uniform, some concern one distinguished physical vector, and some select a sector or block after seeing the vector. Their quantifiers cannot be silently exchanged.
2. Some maps are mandatory parts of the physical transformation, while other projections, singleton floors, and mass-degree estimates are only sufficient certificates. Failure of a certificate does not imply failure of the physical map.
3. The endpoint allows a total fixed-power loss strictly below $1/400$. A numerical difference used inside one certificate is not automatically an additive loss, and the same exponent may not be charged twice.

The purpose of this paper is to put these distinctions into a theorem-level closure calculus. It is both a synthesis and a new finite-dimensional layer. The synthesis records the literal dependency graph. The new layer specializes classical shorted-operator theory to the exact master transfer constant and nuisance spaces of this closure problem, and adds a strict selector-quantifier obstruction, a partial order among reassembly certificates, and an inverse theorem localizing every large missing-alien eigenvalue to a row and a literal determinant cell.

1.1 The master constant

Let $\mathcal{H}_X$ be the original coefficient space, let $0 \neq R_X \succeq 0$ represent the parent energy to be preserved, and let $F_X : \mathcal{H}_X \rightarrow \mathcal{Y}_X$ be the complete declared output map after every mandatory transformation in scope. Define

$$\alpha_X := \inf_{\langle R_X z, z \rangle > 0} \frac{\|F_X z\|^2}{\langle R_X z, z \rangle}. \quad (1)$$

If $R_X = 0$, the quotient has no admissible vector and the transfer problem is vacuous; throughout the master theorem no eigenvalue or loss exponent is assigned to that case. Write

$$N_X := \text{Ker } R_X, \quad S_X := N_X^\perp, \quad W_X := F_X(N_X). \quad (2)$$

The vectors in N_X cost no parent energy, but their outputs can cancel the output of a vector in S_X . Therefore the correct numerator is not the compression $F_{X,S}^* F_{X,S}$. It is the shorted form

$$F_{X,S}^* P_{W_X^\perp} F_{X,S}. \quad (3)$$

We prove

$$\alpha_X = \lambda_{\min} \left(R_{X,S}^{-1/2} F_{X,S}^* P_{W_X^\perp} F_{X,S} R_{X,S}^{-1/2} \right), \quad (4)$$

and the exact finite-scale criterion

$$\alpha_X > 0 \iff \text{Ker } F_X \subseteq \text{Ker } R_X. \quad (5)$$

This extends the TPC–60 reassembly shorting from one represented/missing split to the entire declared packet. It does not assert that α_X has a polynomial lower bound for the arithmetic family.

1.2 Quantifier order is a mathematical gate

For a finite family of selector maps S_a , two superficially similar constants are

$$\eta_{\text{frozen}} := \sup_a \inf_{z \neq 0} \frac{\|S_a z\|^2}{\|z\|^2}, \quad (6)$$

$$\eta_{\text{adaptive}} := \inf_{z \neq 0} \sup_a \frac{\|S_a z\|^2}{\|z\|^2}. \quad (7)$$

Always $\eta_{\text{frozen}} \leq \eta_{\text{adaptive}}$, and equality can fail maximally. Coordinate projections have a positive adaptive constant, although every fixed projection has a nontrivial kernel and hence zero uniform constant. This is the precise obstruction behind the implication

$$\forall z \exists a = a(z) \not\Rightarrow \exists a \forall z.$$

It applies to every coefficient-adaptive sector or maximal block. A legal uniformization retains all selector labels by orthogonal direct sum; summing those outputs back into one physical coordinate is another alias/reassembly gate.

1.3 From a global eigenvalue to a literal obstruction

TPC–61 writes the missing native map after raw diagonal normalization as a matrix F_M with unit columns. Its Gram is

$$K_M = F_M^* F_M = I + \Xi_{\neq p}, \quad \kappa_M = \lambda_{\max}(K_M). \quad (8)$$

The off-diagonal entries are weighted correlations of

$$\mu(mj + h_0) \mu(m'j + h_0), \quad (9)$$

equivalently signed sums on

$$m'pr - mp'r' = h_0(m' - m). \quad (10)$$

The operator norm is a global object, so a single pair estimate is not enough. Conversely, a large operator norm cannot remain completely delocalized. If $\kappa_M \geq 1 + L$, the Schur bound forces one row to have off-diagonal absolute mass at least L . Partitioning that row into B literal correlation cells forces one cell to carry at least L/B . If that cell contains at most Δ neighbors, one normalized determinant sum has size at least $L/(B\Delta)$. This is an inverse localization theorem: it produces a concrete witness to a large alias norm, but no cancellation estimate.

At exponent level, a row-overlap degree $X^{\delta+o(1)}$ and a uniform pair bound $X^{-\tau+o(1)}$ imply

$$\nu_M := \limsup_{X \rightarrow \infty} \frac{\log \max\{1, \kappa_M\}}{\log X} \leq (\delta - \tau)_+. \quad (11)$$

The TPC–61 effective-product door would then open under

$$\sigma_M > \lambda_{\text{rep}} + (\delta - \tau)_+. \quad (12)$$

Neither δ nor τ is favorably estimated here on the arithmetic carrier.

1.4 Main conclusions and scope

The paper proves the following.

1. The master shorted-Gram formula (4) and kernel criterion (5) are exact.
2. Shorting is transitive on nested retained subspaces. A sequence of genuine nuisance eliminations is therefore equivalent to one master elimination, whereas ordinary compression may overstate transfer.
3. A mandatory map with a kernel direction of positive parent energy kills the route exactly. A zero singleton, selection, mass-degree, or other sufficient-certificate constant kills only that certificate unless the final map actually factors through it.
4. Adaptive and frozen selector constants obey a strict min-max gap. Orthogonal direct sum gives the exact uniformization on a labeled output space.
5. Exact shorting dominates every valid reverse-triangle certificate; the restricted missing Gram dominates its residual and ambient relaxations. Residual and ambient relaxations need not dominate one another.
6. Every large missing-alias eigenvalue has a row/cell/pair witness, with the quantitative and exponent consequences above.
7. If represented-chain, reassembly, and downstream lower bounds are proved on the same carrier and in the same quantifier scope, their exponents add. A sum strictly below $1/400$ closes only the declared truncated fixed- h_0 packet lower square.

The first six conclusions and the abstract multiplication statement in the seventh are L0 finite-dimensional theorems. The literal TPC coordinate identification in the sixth conclusion, including (9)–(10), is L1. The fixed- h_0 application of the seventh conclusion is conditional L1. No theorem here proves $\kappa_M = X^{o(1)}$, a new Möbius correlation estimate, a lower square for one fixed prime pair, or any twin-prime statement.

The paper is organized as follows. Section 2 records the logical graph and quantifier signatures. Sections 3 and 4 develop the master closure calculus. Sections 5 and 6 prove the quantifier gap and certificate partial order. Section 7 localizes the missing-alias obstruction. The conditional closure theorem, independence models, endpoint decisions, and next targets appear in sections 8 to 11.

2 The literal dependency graph and quantifier signatures

Historical paper numbers record discovery order, not a license to multiply all displayed constants. This section extracts the logical order needed by a lower-bound proof and assigns a quantifier signature to every arrow.

2.1 Four signatures

Let Z_X be a coefficient class fixed before any vector is chosen. Whenever a uniform generalized eigenvalue or kernel equivalence is used, the declared class is understood to be its full linear coefficient space, or a degree-zero direction-complete class containing a nonzero scalar multiple of every direction. For a proper nonlinear subset, the infimum must instead be restricted to that subset, and a full-space kernel vector outside the subset does not kill the restricted statement.

Definition 2.1 (Quantifier signature). A lower statement has one of the following signatures.

- (i) U (uniform): one fixed carrier and one fixed map satisfy the bound for every $z \in Z_X$;

- (ii) **D** (distinguished): the bound is asserted only for one declared physical vector $z_*(X)$;
- (iii) **A** (adaptive): for every $z \in Z_X$, at least one selector $a = a(z)$ from a fixed family gives the bound;
- (iv) **G** (averaged): only an average over vectors, shifts, selectors, or another external parameter is bounded.

The carrier, map, atomic normalization, and reference form are part of the signature.

The signature is not a stylistic label. It determines which implications are valid.

Proposition 2.2 (Safe signature rules). *The following rules hold.*

- (a) *A **U** statement may be evaluated at z_* , producing a **D** statement on the same carrier and map.*
- (b) *Finitely many sequential **U** inequalities compose when the output form of each arrow is literally the input form of the next.*
- (c) *Sequential **D** inequalities compose only when every arrow concerns the actual intermediate image generated by the same initial vector.*
- (d) *Neither **D**, **A**, nor **G** implies **U** without an additional theorem.*
- (e) *An **A** statement composes as a labeled direct-sum statement if all selector outputs are retained orthogonally. It does not automatically compose after their labels are erased.*
- (f) *A kernel or generalized-eigenvalue statement on the full coefficient space applies to a restricted declared class as a lower certificate, but a kernel witness kills that restricted class only when the witness direction belongs to its homogeneous cone.*

Proof. Parts (a)–(c) are direct substitutions with their quantifiers unchanged. The coordinate-projection example in [section 5](#) proves the failure in part (d) for **A**; one-vector and averaging examples prove the other two failures. Part (e) is the direct-sum identity proved in [theorem 5.3](#); erasing labels applies a further linear map and can create cancellation. Part (f) follows because an infimum over the full space is no larger than the infimum over a subset, whereas a zero quotient obstructs a subset only if that subset contains the corresponding direction up to nonzero scaling. $\square$

Changing one fixed nonzero shift into a shift average changes both the carrier and the signature. Likewise, dividing an atomic input norm by a fiber size, choosing a coefficient-dependent support, or replacing a uniform generalized eigenvalue by one Rayleigh quotient changes the statement rather than proving it.

2.2 The logical graph

The fixed-shift route has the following logical skeleton:

$$\begin{aligned}
& \text{literal parent atoms} \longrightarrow \text{actual-energy selection} \longrightarrow \text{critical high-face activation} \\
& \longrightarrow \text{terminal prime-resolved fibers} \longrightarrow \text{represented residual/native output} \\
& \longrightarrow \text{represented plus missing reassembly} \longrightarrow \text{boundary/tail reconnection.}
\end{aligned} \tag{13}$$

Each phrase abbreviates several exact objects. The status inherited from TPC–53–TPC–61 is summarized below.

Source	Logical object	Established conclusion	Closure status
TPC-53	Literal factor dictionary	Canonical row/orbit/prime/cofactor coordinates, literal coefficients, raw atomic normalization, and an interior compatible cell.	L1 interface
TPC-54	Actual-energy selection	Sharp carrier-coverage and saturation laws. Cofactor occupancy and a coefficient floor remain inputs.	L0/L1 certificate
TPC-59	Critical high face	The admissible cutoff may be $y \asymp Q$; exact occupancy, activation, and dyadic localization constants are identified.	L1 route correction
TPC-55	Prime-resolved terminal fibers	Exact grouping SVD, common Möbius gauge, pseudoinverse, and collision-free transversal on the declared terminal support.	L0/L1 structure
TPC-56	Phase-sector extraction	A constant-energy sector exists after a coefficient-adaptive choice. The sector is not one frozen linear output for all coefficients.	A, not U
TPC-57	Undeleted singleton route	Exact collision moments and the sharp singleton floor. A positive endpoint singleton fraction is not proved.	U certificate
TPC-58	Residual/native affine map	Exact erasure quotient, dark-space test, and two-stage generalized Grams. No endpoint lower constant is obtained.	L0/L1 gate
TPC-60	High-face reassembly	Exact represented/missing split, shorted Gram, kernel test, and mass-degree certificate.	L0/L1 gate
TPC-61	Missing cofactor exposure and alias Gram	Exact exposure measure, cofactor ladder, restricted κ_M , and the two-affine Möbius/determinant kernel.	L0/L1 reduction

The table is not a proof that all rows are simultaneously active. In particular, actual-energy selection does not prove critical terminal activation; prime-resolved isometry does not prove erasure injectivity; singleton incidence does not prove a positive singleton mass; and an exact formula for the missing alias Gram does not bound its top eigenvalue.

2.3 Branches that must not be spliced

The terminal-prime layer has two useful but logically different branches.

Adaptive phase branch. TPC-56 proves that after inspecting a coefficient vector one can retain a phase sector with a constant fraction of the already grouped fiber energy [6]. This is an A statement. If the goal is a coefficient-uniform lower bound for one undeleted physical map, the selected sector cannot be inserted as a fixed arrow. It becomes uniform only on a new labeled direct-sum output, after which reassembly to the physical output is still open.

Undeleted singleton branch. TPC-57 stays on the undeleted output and gives an exact sufficient floor in terms of singleton energy [7]. This branch has the right uniform carrier, but the singleton floor may vanish. Such vanishing kills that certificate, not the complete coherent output, because nonsingleton fibers may still add favorably.

Thus the two branches cannot be combined by taking the positive part of each: the first changes the selector with the vector, while the second keeps the map fixed but may have zero certified mass.

2.4 TPC-59 is logically upstream of the terminal endpoint

TPC-58 inherited a cutoff displaced by a fixed positive power. TPC-59 shows that the literal very-high conditions already permit

$$y = C_{\text{vh}}Q \quad (14)$$

for a sufficiently large support-dependent constant, reopening the critical cofactor scale

$$R \asymp J = X^{133/400+o(1)}. \quad (15)$$

Accordingly, TPC-59 is historically later than TPC-55–TPC-58 but logically upstream of any endpoint use of their terminal carrier. The cutoff correction is a feasibility statement. It does not prove that the actual retained relation occupies the reopened block.

2.5 The surviving three modules

After exact identities and zero-cost coordinate changes have been removed, the prospective lower chain can be compressed into three modules.

Module I. The *represented chain*: actual selection, critical occupancy, terminal activation, and represented residual/native transfer produce a represented map A_X .

Module II. The *reassembly gate*: the missing map M_X is restored and $A_X + M_X$ is compared with A_X , uniformly by shorting or in a declared distinguished scope.

Module III. The *downstream reconnection*: boundary, tail, and parent-target interfaces compare the complete slice output with the declared truncated packet target.

The conditional theorem in [section 8](#) composes these modules only after their carriers, normalizations, and signatures have been matched.

2.6 Six-item physical audit

Every L1 specialization in this paper is subject to the following audit.

- (A1) *Literal coefficients.* Atom amplitudes are the inherited physical coefficients; no favorable atom phases are chosen.
- (A2) *Fixed shift.* Every target is built from the same fixed nonzero h_0 , with no shift average substituted.
- (A3) *Raw atomic norm.* The input energy is raw counting measure, not a fiber-normalized surrogate.
- (A4) *Canonical representation.* Row, orbit, largest-prime, cofactor, residual, and native labels are recovered canonically; only identically inactive directions are removed.
- (A5) *Actual support.* Carriers and selector families are fixed before the coefficient vector, unless the conclusion is explicitly labeled adaptive.

(A6) *Endpoint budget.* Structural identities cost zero; every proved lower loss is entered once in a strict total below 1/400.

Passing this audit says that the correct physical object and quantifier have been retained. It does not supply any of the open arithmetic lower bounds. Here h_0 remains an arbitrary fixed nonzero shift. Nothing in this audit specializes it to $h_0 = 2$, and no conclusion for that special shift may be inferred without the missing arithmetic and downstream identifications.

3 The master shorted–Gram gate

The shorted-operator construction and its Loewner-maximal characterization are classical, originating with Anderson and the Anderson–Trapp development [1, 2]. We include the finite-dimensional proof to fix the exact quotient, nuisance space, and quantifiers needed by the TPC closure specialization; no priority claim is made for the general shorted-operator theory.

This section isolates the finite-dimensional closure problem before any arithmetic specialization. Let $\mathcal{K}$ and $\mathcal{H}$ be finite-dimensional complex Hilbert spaces, let

$$R = R^* \geq 0 \quad \text{on } \mathcal{K}, \quad F : \mathcal{K} \longrightarrow \mathcal{H}, \quad (16)$$

and assume $R \neq 0$. The parent form is $z \mapsto \langle Rz, z \rangle$, while F is the final physical map after every declared layer has been assembled. Define the exact closure constant

$$\alpha_R(F) := \inf_{\langle Rz, z \rangle > 0} \frac{\|Fz\|^2}{\langle Rz, z \rangle}. \quad (17)$$

We also write $\alpha(F; R) := \alpha_R(F)$ when the map is the first argument in a chain of certificates. The order of the quantifiers in (17) is part of the definition: one constant must control every parent-visible vector. The exclusion $R = 0$ is substantive rather than cosmetic. In that case there is no parent-visible vector, the infimum in (17) has empty domain, and the least eigenvalue on $S = \{0\}$ is not defined. Every inequality $\|Fz\|^2 \geq c\langle Rz, z \rangle$ is then vacuous, so the zero parent form is treated separately and is assigned no closure constant.

Put

$$N := \text{Ker } R, \quad S := N^\perp = \text{Ran } R, \quad R_S := R|_S, \quad (18)$$

so that R_S is positive definite on the nonzero space S . Write

$$F_S := F|_S, \quad F_N := F|_N, \quad W := F(N) = \text{Ran } F_N, \quad P := P_{W^\perp}. \quad (19)$$

The subspace W is important: vectors in N cost no parent energy, but their final images may cancel the image of a parent-visible vector.

3.1 The finite-dimensional short

For a positive semidefinite operator G on $\mathcal{K}$ and a subspace $L \subseteq \mathcal{K}$, its *short to* L , denoted $\text{Sh}_L(G)$, is the unique positive operator characterized by

$$\text{Sh}_L(G) = \max\{B : 0 \leq B \leq G, \text{Ran } B \subseteq L\} \quad (20)$$

in the Loewner order. For completeness, it has the following exact variational description. If $s \in L$, then

$$\langle \text{Sh}_L(G)s, s \rangle = \min_{n \in L^\perp} \langle G(s+n), s+n \rangle, \quad (21)$$

and $\text{Sh}_L(G)$ vanishes on $L^\perp$.

To verify both existence and maximality without invoking an infinite-dimensional theorem, write $G = C^*C$ and put $V = C(L^\perp)$. Least squares gives

$$\min_{n \in L^\perp} \|Cs + Cn\|^2 = \|P_{V^\perp}Cs\|^2. \quad (22)$$

The right-hand side is a positive quadratic form on L , and therefore defines a positive operator supported on L . Call it B_L . For $z = s + n$, (21) gives $\langle B_L z, z \rangle \leq \langle Gz, z \rangle$, so $B_L \leq G$. Conversely, if $0 \leq B \leq G$ and $\text{Ran } B \subseteq L$, then $B = P_L B P_L$, and for every $n \in L^\perp$,

$$\langle Bs, s \rangle = \langle B(s + n), s + n \rangle \leq \langle G(s + n), s + n \rangle.$$

Taking the minimum over n shows $B \leq B_L$. Hence $B_L = \text{Sh}_L(G)$, proving (20) and (21) simultaneously.

Apply this construction to the final Gram operator

$$G := F^*F. \quad (23)$$

Relative to $\mathcal{K} = S \oplus N$, write

$$G = \begin{pmatrix} G_{SS} & G_{SN} \\ G_{NS} & G_{NN} \end{pmatrix} = \begin{pmatrix} F_S^* F_S & F_S^* F_N \\ F_N^* F_S & F_N^* F_N \end{pmatrix}. \quad (24)$$

Lemma 3.1 (Projected Gram and generalized Schur complement). *The short of G to S is*

$$\text{Sh}_S(G) = \begin{pmatrix} Q & 0 \\ 0 & 0 \end{pmatrix}, \quad \boxed{Q = F_S^* P F_S}. \quad (25)$$

Equivalently,

$$Q = G_{SS} - G_{SN} G_{NN}^\dagger G_{NS}. \quad (26)$$

For each $s \in S$,

$$\min_{n \in N} \|F_S s + F_N n\|^2 = \|P F_S s\|^2 = \langle Qs, s \rangle. \quad (27)$$

All minimizers are

$$n_s = -F_N^\dagger F_S s + n_0, \quad n_0 \in \text{Ker } F_N. \quad (28)$$

Proof. The first equality in (27) is the least-squares distance from $-F_S s$ to $W = \text{Ran } F_N$. The Moore–Penrose least-squares formula gives (28). Consequently the quadratic form of the short is $\|P F_S s\|^2$, proving (25). Finally,

$$F_N (F_N^* F_N)^\dagger F_N^* = P_W,$$

and substitution in the block expression gives (26). $\square$

3.2 The master eigenvalue and exact kernel decision

Theorem 3.2 (Master shorted–Gram formula). *For the data in (16),*

$$\boxed{\alpha_R(F) = \lambda_{\min}(R_S^{-1/2} Q R_S^{-1/2} \big|_S), \quad W = F(\text{Ker } R), \quad Q = F_S^* P_{W^\perp} F_S.} \quad (29)$$

Equivalently,

$$\alpha_R(F) = \lambda_{\min}(R^{\dagger/2} \text{Sh}_S(F^* F) R^{\dagger/2} \big|_S). \quad (30)$$

Moreover,

$$\boxed{\alpha_R(F) > 0 \iff \text{Ker } F \subseteq \text{Ker } R.} \quad (31)$$

Thus failure of the kernel inclusion is an exact finite-scale kill, not a loss in a sufficient estimate.

Proof. Every $z \in \mathcal{K}$ has a unique decomposition $z = s + n$, with $s \in S$ and $n \in N$. Since $Rn = 0$,

$$\langle Rz, z \rangle = \langle R_S s, s \rangle,$$

which is positive precisely when $s \neq 0$. Minimizing the numerator over n and using [lemma 3.1](#) gives

$$\alpha_R(F) = \inf_{s \in S \setminus \{0\}} \frac{\langle Qs, s \rangle}{\langle R_S s, s \rangle}. \quad (32)$$

The substitution $x = R_S^{1/2} s$ and the Rayleigh–Ritz principle prove [\(29\)](#). On S , $R^{\dagger/2} = R_S^{-1/2}$, while both $R^{\dagger/2}$ and $\text{Sh}_S(F^*F)$ vanish on N ; this proves [\(30\)](#).

If $\text{Ker } F \not\subseteq \text{Ker } R$, choose $z \in \text{Ker } F$ with $\langle Rz, z \rangle > 0$. Its quotient is zero, so $\alpha_R(F) = 0$. Conversely, suppose $\text{Ker } F \subseteq N$. If $PF_S s = 0$, then $F_S s \in W$, so there is $n \in N$ with $F_S s + F_N n = 0$. Hence $s + n \in \text{Ker } F \subseteq N$. Projecting this vector orthogonally onto S gives $s = 0$. Thus PF_S is injective on S . In finite dimension its least generalized singular value relative to the positive definite form R_S is strictly positive. Equation [\(29\)](#) now gives $\alpha_R(F) > 0$. $\square$

Corollary 3.3 (Dimension kill). *If*

$$\dim S > \dim \mathcal{H} - \dim W, \quad (33)$$

then $\alpha_R(F) = 0$.

Proof. The effective map $PF_S : S \rightarrow W^\perp$ cannot be injective under [\(33\)](#); apply [\(31\)](#). $\square$

The converse to [corollary 3.3](#) is false: compatible dimensions do not force injectivity. Even the exact kernel inclusion has no asymptotic size content. For example, with $R_X = I$ and $F_X = e^{-X}I$, one has $\text{Ker } F_X = \{0\}$ and

$$\alpha_{R_X}(F_X) = e^{-2X} > 0 \quad (34)$$

at every finite scale, but there is no finite exponent λ for which $\alpha_{R_X}(F_X) \geq X^{-\lambda+o(1)}$. Kernel survival and an endpoint-quality polynomial certificate are therefore separate gates.

3.3 A mandatory factor: an actual kill versus a failed certificate

Suppose the final map has a genuine factorization

$$\mathcal{K} \xrightarrow{T} \mathcal{Y} \xrightarrow{L} \mathcal{H}, \quad F = LT. \quad (35)$$

Calling T *mandatory* means the identity [\(35\)](#) holds for the literal final map; it does not mean merely that one proof happens to use T .

Proposition 3.4 (Mandatory-factor kernel kill). *If*

$$\text{Ker } T \not\subseteq \text{Ker } R, \quad (36)$$

then

$$\alpha_R(LT) = 0 \quad (37)$$

for every downstream map L . This conclusion requires an actual vector in $\text{Ker } T \setminus \text{Ker } R$; failure to prove a positive bound for T , or failure of one factorwise sufficient certificate, is not such a vector.

Proof. Choose $z \in \text{Ker } T \setminus \text{Ker } R$. Positivity of R implies $\langle Rz, z \rangle > 0$, whereas $LTz = 0$. Thus the quotient in [\(17\)](#) vanishes. Equivalently, $\text{Ker } T \subseteq \text{Ker}(LT)$, so the kernel criterion [\(31\)](#) fires. $\square$

The converse of [proposition 3.4](#) is deliberately not asserted. The inclusion $\text{Ker } T \subseteq \text{Ker } R$ says only that the mandatory factor itself has not killed a parent-visible vector. A downstream map can still do so: for $R = I$, $T = I$, and $L = 0$, the inclusion holds while $\alpha_R(LT) = 0$.

Here is a precise sufficient certificate whose failure illustrates the logical distinction. When $T \neq 0$, define

$$\tau_R(T) := \inf_{\langle Rz, z \rangle > 0} \frac{\|Tz\|^2}{\langle Rz, z \rangle}, \quad (38)$$

$$\ell(L; T) := \inf_{0 \neq y \in \text{Ran } T} \frac{\|Ly\|^2}{\|y\|^2}. \quad (39)$$

Then, directly from the two definitions,

$$\alpha_R(LT) \geq \ell(L; T) \tau_R(T). \quad (40)$$

The equality $\tau_R(T) = 0$ is an exact mandatory-factor kill precisely when $\text{Ker } T \not\subseteq \text{Ker } R$, by [theorem 3.2](#) applied to T . In contrast, $\ell(L; T) = 0$ may only say that L has a kernel on a direction which the parent form already regards as invisible.

Example 3.5 (A factorwise certificate can fail while closure is exact). Take $\mathcal{K} = \mathcal{Y} = \mathbb{C}^2$, $\mathcal{H} = \mathbb{C}$, and

$$R = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad T = I, \quad L(x, y) = x. \quad (41)$$

Here $\tau_R(T) = 1$, but $\ell(L; T) = 0$ because $L(0, 1) = 0$. Thus the right-hand side of (40) is zero. Nevertheless

$$\frac{\|LT(x, y)\|^2}{\langle R(x, y), (x, y) \rangle} = 1 \quad (x \neq 0), \quad (42)$$

so $\alpha_R(LT) = 1$. Indeed $\text{Ker}(LT) = \text{span}\{(0, 1)\} = \text{Ker } R$. The certificate failed; the gate did not.

At an asymptotic scale, the same distinction must be maintained. An exhibited vector in $\text{Ker } T_X \setminus \text{Ker } R_X$ is a rigorous zero at that scale. A nonpositive exponent budget, an uncontrolled norm, or the absence of a proof of $\tau_{R_X}(T_X) \gg X^{-C}$ is only a failure of the chosen route. It cannot be promoted to an operator kernel statement.

4 Nested gates and exact sequential elimination

Transitivity for nested shorts is part of the classical shorted-operator calculus [1, 2]. The self-contained proof below records precisely how that identity acts on the staged nuisance spaces of the present closure problem.

The master formula eliminates the full parent-null space at once. A proof architecture may instead expose nuisance directions in several layers. This section shows that exact shorting is transitive along nested retained spaces, so the sequential and one-shot procedures agree. The statement is purely finite-dimensional and makes no arithmetic assertion.

4.1 Transitivity of the short

Theorem 4.1 (Transitivity on nested retained spaces). *Let $G \geq 0$ on a finite-dimensional Hilbert space $\mathcal{K}$, and let*

$$T \subseteq S \subseteq \mathcal{K} \quad (43)$$

be subspaces. Then

$$\boxed{\text{Sh}_T(\text{Sh}_S(G)) = \text{Sh}_T(G)}. \quad (44)$$

Proof. Use the maximal characterization (20). The operator $\text{Sh}_T(\text{Sh}_S(G))$ is supported on T and satisfies

$$0 \leq \text{Sh}_T(\text{Sh}_S(G)) \leq \text{Sh}_S(G) \leq G.$$

It is therefore admissible in the defining maximum for $\text{Sh}_T(G)$, and hence

$$\text{Sh}_T(\text{Sh}_S(G)) \leq \text{Sh}_T(G). \quad (45)$$

Conversely, $\text{Sh}_T(G)$ is supported on $T \subseteq S$ and is bounded above by G . It is therefore admissible in the defining maximum for $\text{Sh}_S(G)$, so $\text{Sh}_T(G) \leq \text{Sh}_S(G)$. Since it is supported on T , it is then admissible in the defining maximum for $\text{Sh}_T(\text{Sh}_S(G))$. Thus

$$\text{Sh}_T(G) \leq \text{Sh}_T(\text{Sh}_S(G)). \quad (46)$$

Combining the two inequalities proves (44). $\square$

There is also a direct variational reading. Since $T \subseteq S$,

$$T^\perp = (S \cap T^\perp) \oplus S^\perp. \quad (47)$$

For $t \in T$, applying (21) twice gives

$$\begin{aligned} \langle \text{Sh}_T(\text{Sh}_S(G))t, t \rangle &= \min_{u \in S \cap T^\perp} \min_{n \in S^\perp} \langle G(t + u + n), t + u + n \rangle \\ &= \min_{v \in T^\perp} \langle G(t + v), t + v \rangle \\ &= \langle \text{Sh}_T(G)t, t \rangle. \end{aligned} \quad (48)$$

Thus transitivity is exactly the associativity of eliminating independent orthogonal nuisance layers. Nestedness is essential to this statement; no commutation claim is made for two unrelated retained spaces.

4.2 A ladder of nuisance spaces

Let

$$\mathcal{K} = S_0 \supseteq S_1 \supseteq \cdots \supseteq S_k \quad (49)$$

be a nested ladder, and define the nuisance layer removed at stage j by

$$E_j := S_{j-1} \cap S_j^\perp, \quad 1 \leq j \leq k. \quad (50)$$

Then

$$S_{j-1} = S_j \oplus E_j, \quad S_j^\perp = E_j \oplus E_{j-1} \oplus \cdots \oplus E_1 \quad (1 \leq j \leq k). \quad (51)$$

Let $F : \mathcal{K} \rightarrow \mathcal{H}$, set $G_0 = F^*F$, and recursively define

$$G_j := \text{Sh}_{S_j}(G_{j-1}), \quad 1 \leq j \leq k. \quad (52)$$

Theorem 4.2 (Sequential elimination equals one shot). *For every j ,*

$$\boxed{G_j = \text{Sh}_{S_j}(F^*F)}. \quad (53)$$

Moreover, for $s \in S_j$,

$$\langle G_j s, s \rangle = \min_{e_j \in E_j} \min_{e_{j-1} \in E_{j-1}} \cdots \min_{e_1 \in E_1} \left\| F \left(s + \sum_{i=1}^j e_i \right) \right\|^2 \quad (54)$$

$$= \min_{n \in S_j^\perp} \|F(s + n)\|^2. \quad (55)$$

All minima are attained.

Proof. The operator identity follows inductively from [theorem 4.1](#):

$$G_j = \text{Sh}_{S_j}(\text{Sh}_{S_{j-1}}(F^*F)) = \text{Sh}_{S_j}(F^*F).$$

For the form identity, the variational formula for shorting from S_{j-1} to S_j is

$$\langle G_j s, s \rangle = \min_{e_j \in E_j} \langle G_{j-1}(s + e_j), s + e_j \rangle. \quad (56)$$

Iterating (56) proves (54). The orthogonal direct sum (51) identifies $\sum_{i=1}^j e_i$ with an arbitrary vector of $S_j^\perp$, proving (55). Every space is finite dimensional, and the least-squares argument (22) shows that the minima are attained. $\square$

For a nonzero parent form in [section 3](#), take the first retained space to be

$$S_1 = (\text{Ker } R)^\perp. \quad (57)$$

The first eliminated layer is $E_1 = \text{Ker } R$, and $G_1 = \text{Sh}_{S_1}(F^*F)$ is precisely the numerator in (30). If a subsequent argument has genuinely nested retained spaces $S_k \subseteq \cdots \subseteq S_1$, then

$$\text{Sh}_{S_k}(\cdots \text{Sh}_{S_2}(\text{Sh}_{S_1}(F^*F)) \cdots) = \text{Sh}_{S_k}(F^*F). \quad (58)$$

Consequently no extra loss is created merely by staging exact nuisance elimination. Any loss appearing only after replacing a short by an upper or lower certificate belongs to that certificate, not to the underlying finite-dimensional elimination.

4.3 Why compression is not elimination

Ordinary compression to S is

$$\mathcal{C}_S(G) := P_S G P_S. \quad (59)$$

It evaluates the form after setting the discarded coefficient to zero. Shorting instead minimizes over that coefficient. Therefore

$$0 \leq \text{Sh}_S(G) \leq \mathcal{C}_S(G) \quad (60)$$

as operators supported on S , and equality need not hold. The next example shows that one compression in a nested chain can turn an exact zero into a false positive constant.

Example 4.3 (Intermediate compression destroys exact cancellation). *Let $\mathcal{K} = \mathbb{C}^3$, $\mathcal{H} = \mathbb{C}^2$, and*

$$F(x, y, z) = (x + y, y + z). \quad (61)$$

Take

$$S_1 = \text{span}\{e_1, e_2\}, \quad S_2 = \text{span}\{e_1\}. \quad (62)$$

The Gram matrix is

$$G = F^*F = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}. \quad (63)$$

Exact elimination of z gives, on S_1 ,

$$\min_{z \in \mathbb{C}} (|x + y|^2 + |y + z|^2) = |x + y|^2, \quad (64)$$

and exact elimination of y then gives zero. Equivalently,

$$\text{Sh}_{S_2}(\text{Sh}_{S_1}(G)) = \text{Sh}_{S_2}(G) = 0, \quad (65)$$

because for any x one may choose $y = -x$ and $z = x$, making $F(x, y, z) = 0$.

If the first short is incorrectly replaced by compression, the form on S_1 is instead

$$|x + y|^2 + |y|^2. \quad (66)$$

Shorting this compressed form to S_2 yields

$$\min_{y \in \mathbb{C}} (|x + y|^2 + |y|^2) = \frac{1}{2}|x|^2, \quad (67)$$

where the minimizer is $y = -x/2$. Thus

$$\text{Sh}_{S_2}(\mathcal{C}_{S_1}(G)) = \frac{1}{2}P_{S_2} \neq 0. \quad (68)$$

Compressing at both stages is still worse:

$$\mathcal{C}_{S_2}(\mathcal{C}_{S_1}(G)) = P_{S_2}. \quad (69)$$

The numbers 0, $1/2$, and 1 therefore arise from, respectively, exact elimination, one premature compression, and two premature compressions of the same positive Gram matrix.

The mechanism is not a numerical pathology. In the example, the discarded coordinate z cancels the second output component. Setting $z = 0$ retains a spurious penalty $|y|^2$, which prevents the next layer from using $y = -x$ and manufactures a positive lower bound. In a physical closure argument, a coefficient direction may be compressed away only after proving that it cannot alter the final output, or after replacing the desired statement by one explicitly restricted to the compressed coefficient class.

4.4 Exact decisions and route failures

Combining [theorems 3.2](#) and [4.2](#) gives a quantifier-safe decision rule. After all genuine nuisance layers have been shorted, an actual zero eigenvalue is equivalent to a final kernel vector carrying positive parent energy. It kills the uniform closure gate on the declared class. By contrast, any of the following is only a failure of a proposed proof route:

- (i) replacing a shorted operator by a compression and obtaining an unusable bound;
- (ii) failing to control a sufficient singular-value or norm certificate;
- (iii) exhausting an exponent budget in an upper-bound calculation;
- (iv) proving a pointwise estimate for one distinguished vector but not the uniform generalized eigenvalue.

None of these statements supplies a vector in $\text{Ker } F \setminus \text{Ker } R$. Conversely, the mandatory-factor witness in [proposition 3.4](#) does supply such a vector and therefore survives every reformulation of the certificate. This is the strict L0 boundary used below: exact operator kernels decide impossibility of the declared uniform gate, whereas failed certificates decide only which route should be abandoned or redesigned.

5 Adaptive selectors and the quantifier barrier

Several earlier TPC reductions locate, for each coefficient vector, a sector, block, or face carrying a definite portion of its mass. Such a statement has the quantifier order

$$\text{for every } z \text{ there exists a label } a = a(z).$$

It does not by itself provide one label which works for every coefficient vector. This section isolates that distinction as finite-dimensional linear algebra. No arithmetic input is used.

5.1 The two selector constants

Let $\mathcal{K} \neq \{0\}$ be a finite-dimensional complex Hilbert space, let $\mathcal{A}$ be a finite nonempty label set, and let

$$S_a : \mathcal{K} \longrightarrow \mathcal{H}_a \quad (a \in \mathcal{A}) \quad (70)$$

be linear maps into finite-dimensional Hilbert spaces. For $z \neq 0$, put

$$q_a(z) := \frac{\|S_a z\|^2}{\|z\|^2}. \quad (71)$$

The adaptive and fixed-label lower constants are respectively

$$\lambda_{\text{ad}} := \inf_{z \neq 0} \max_{a \in \mathcal{A}} q_a(z), \quad (72)$$

$$\lambda_{\text{fix}} := \max_{a \in \mathcal{A}} \inf_{z \neq 0} q_a(z) = \max_{a \in \mathcal{A}} \lambda_{\min}(S_a^* S_a). \quad (73)$$

The extrema exist because the unit sphere is compact and $\mathcal{A}$ is finite.

Theorem 5.1 (Quantifier-safe minimax inequality). *For every family (70),*

$$\boxed{\sup_{a \in \mathcal{A}} \inf_{z \neq 0} q_a(z) \leq \inf_{z \neq 0} \sup_{a \in \mathcal{A}} q_a(z).} \quad (74)$$

Equivalently, $\lambda_{\text{fix}} \leq \lambda_{\text{ad}}$. Consequently, for any $c > 0$,

$$\lambda_{\text{ad}} \geq c \iff (\forall z \neq 0)(\exists a \in \mathcal{A}) : \|S_a z\|^2 \geq c \|z\|^2, \quad (75)$$

$$\lambda_{\text{fix}} \geq c \iff (\exists a \in \mathcal{A})(\forall z \neq 0) : \|S_a z\|^2 \geq c \|z\|^2. \quad (76)$$

The first statement does not imply the second.

Proof. For every fixed pair (a, z) ,

$$\inf_{w \neq 0} q_a(w) \leq q_a(z) \leq \sup_{b \in \mathcal{A}} q_b(z).$$

Taking the supremum over a in the left member and then the infimum over z in the right member proves (74). The two logical equivalences are merely the definitions of the two constants, together with attainment of the finite maximum and the compact-sphere minima. The failure of the converse is witnessed quantitatively by the next example. $\square$

Example 5.2 (Coordinate projections give a strict gap). *Fix $r \geq 2$, take $\mathcal{K} = \mathbb{C}^r$, $\mathcal{A} = \{1, \dots, r\}$, and let*

$$S_a z = z_a \in \mathbb{C}.$$

For every $z \neq 0$,

$$\max_{1 \leq a \leq r} \frac{|z_a|^2}{\|z\|^2} \geq \frac{1}{r},$$

with equality at $z = r^{-1/2}(1, \dots, 1)$. Hence

$$\lambda_{\text{ad}} = \frac{1}{r}. \quad (77)$$

On the other hand, each S_a has a nonzero kernel, and therefore

$$\lambda_{\text{fix}} = 0. \quad (78)$$

Thus a label can always be chosen after seeing z , but no single label has any positive lower bound on all directions. The gap is not a limiting or measurability artifact: it occurs in a finite family of coordinate projections in finite dimension.

The same example remains valid after multiplying every S_a by any common scale. It therefore obstructs not only a constant lower bound but also any attempted fixed-power upgrade which preserves the same selector family: an adaptive gain $X^{-\lambda}$ does not produce a fixed label with gain $X^{-\lambda+o(1)}$.

5.2 Uniformization by retaining the label

There is a legitimate way to keep all adaptive alternatives without reversing their quantifiers: retain the selector label as an orthogonal output coordinate. Define the labelled analysis map

$$S : \mathcal{K} \longrightarrow \bigoplus_{a \in \mathcal{A}} \mathcal{H}_a, \quad Sz := (S_a z)_{a \in \mathcal{A}}. \quad (79)$$

Theorem 5.3 (Exact labelled direct-sum uniformization). *The optimal coefficient-uniform constant for the label-retaining map is*

$$\lambda_{\text{lab}} := \inf_{z \neq 0} \frac{\|Sz\|^2}{\|z\|^2} = \lambda_{\min} \left(\sum_{a \in \mathcal{A}} S_a^* S_a \right). \quad (80)$$

Moreover,

$$\lambda_{\text{ad}} \leq \lambda_{\text{lab}} \leq |\mathcal{A}| \lambda_{\text{ad}}. \quad (81)$$

In particular, an adaptive bound $\lambda_{\text{ad}} \geq c$ gives the genuine uniform labelled inequality

$$\sum_{a \in \mathcal{A}} \|S_a z\|^2 \geq c \|z\|^2 \quad (z \in \mathcal{K}). \quad (82)$$

Conversely, $\lambda_{\text{lab}} \geq L$ implies only the adaptive bound $\lambda_{\text{ad}} \geq L/|\mathcal{A}|$ without further information.

Proof. Orthogonality of the labelled summands gives

$$\begin{aligned} \|Sz\|^2 &= \sum_{a \in \mathcal{A}} \|S_a z\|^2 \\ &= \left\langle \left(\sum_{a \in \mathcal{A}} S_a^* S_a \right) z, z \right\rangle. \end{aligned}$$

Rayleigh–Ritz proves (80). Pointwise in $z \neq 0$,

$$\max_a q_a(z) \leq \sum_a q_a(z) \leq |\mathcal{A}| \max_a q_a(z).$$

Taking the infimum over z proves (81), and the two consequences follow immediately. $\square$

For the coordinate family in [example 5.2](#), one has

$$\sum_{a=1}^r S_a^* S_a = I_{\mathbb{C}^r}, \quad \lambda_{\text{lab}} = 1,$$

even though $\lambda_{\text{fix}} = 0$. Nothing contradictory has happened: the direct sum retains all r labels, whereas a fixed selector retains only one.

5.3 Label retention is not physical reassembly

The labelled theorem changes the output space. A later map which erases or recombines the label can restore the very cancellations that the direct sum prevented. This is an independent gate.

Proposition 5.4 (Recombination requires a new lower certificate). *Positivity of λ_{lab} alone gives no universal positive coefficient-uniform lower bound after an arbitrary linear recombination*

$$C : \bigoplus_{a \in \mathcal{A}} \mathcal{H}_a \longrightarrow \mathcal{H}.$$

Precisely, there exist a labelled map with $\lambda_{\text{lab}} > 0$ and a recombination C for which CS has lower constant zero. There also exist a bounded-below represented map $A = S$ and a missing map M for which $A + M$ has lower constant zero.

Proof. For the first assertion, take $\mathcal{K} = \mathbb{C}^2$, $S_1(z_1, z_2) = z_1$, $S_2(z_1, z_2) = z_2$, and

$$C(y_1, y_2) = y_1 + y_2.$$

Then $S = I_{\mathbb{C}^2}$ and $\lambda_{\text{lab}} = 1$, while $CS(1, -1) = 0$. Thus the recombined map has lower constant zero.

For the second assertion, on the labelled output space take $A = S$ and $M = -S$. Then A has lower constant $\lambda_{\text{lab}} > 0$, but $A + M = 0$. Hence neither label retention nor a represented lower bound controls signed reassembly without an additional angle, defect, or exact shorted-Gram estimate. $\square$

The correct scope statement is therefore

$$\boxed{\text{adaptive selector} \implies \text{uniform labelled direct sum} \not\Rightarrow \text{uniform physical recombination.}} \quad (83)$$

The first arrow is [theorem 5.3](#); the failed second arrow is [proposition 5.4](#). To pass the second arrow one must analyze the actual physical recombination and, when an undeleted represented map A is perturbed by a missing contribution M , the actual map $A + M$. The exact shorted constant and its certificate lattice are developed in the next section.

5.4 Scope audit

All results in this section are L0 finite-dimensional statements. They make three claims and no others.

- (i) An adaptive selector does not supply a fixed coefficient-uniform selector.
- (ii) Retaining the selector label gives a different, rigorously uniform direct-sum map with exact constant $\lambda_{\min}(\sum_a S_a^* S_a)$.
- (iii) Dropping or recombining that label, or adding a missing signed map, requires a separate reassembly theorem.

In particular, no sector, block, or label identified after inspecting a coefficient vector is promoted here to one physical label valid for all vectors. No estimate for a fixed arithmetic shift, no parity improvement, and no prime-pair lower bound follows from this section.

6 The reassembly certificate lattice

The same physical reassembly map admits several lower certificates. They do not all contain the same information. This section places the exact shorted constant, exact missing-to-represented ratios, the restricted native Gram, a two-stage residual certificate, and the ambient missing degree in one partial order. The order is valid only when the scope—all coefficient vectors or one distinguished vector—is held fixed.

6.1 The exact shorted constant

Let $\mathcal{K}, \mathcal{H}$ be finite-dimensional complex Hilbert spaces and let

$$A, M : \mathcal{K} \longrightarrow \mathcal{H}, \quad B := A + M, \quad A \neq 0. \quad (84)$$

The represented map is A , the missing signed contribution is M , and B is the physical reassembly. Its optimal coefficient-uniform constant relative to the represented norm is

$$\delta_{\text{sh}}(A, M) := \inf_{Az \neq 0} \frac{\|Bz\|^2}{\|Az\|^2}. \quad (85)$$

In the notation of the master gate, this is exactly $\alpha_{A^*A}(A + M)$. We repeat the specialization here so that every node of the certificate lattice is defined in one place.

Put

$$N := \text{Ker } A, \quad S := N^\perp, \quad A_S := A|_S, \quad (86)$$

and write

$$B_S := B|_S, \quad B_N := B|_N = M|_N, \quad P := P_{(\text{Ran } B_N)^\perp}. \quad (87)$$

Finally set

$$R_S := A_S^* A_S > 0, \quad Q_S := B_S^* P B_S \geq 0. \quad (88)$$

Theorem 6.1 (Exact shorted reassembly constant). *For the maps in (84),*

$$\boxed{\delta_{\text{sh}}(A, M) = \lambda_{\min}(R_S^{-1/2} Q_S R_S^{-1/2})}. \quad (89)$$

This is an identity, not a sufficient estimate.

Proof. Every $z \in \mathcal{K}$ has a unique decomposition $z = s + n$, with $s \in S$ and $n \in N$. The denominator depends only on s : $Az = A_S s$. For this fixed s , least squares gives

$$\begin{aligned} \inf_{n \in N} \|B_S s + B_N n\|^2 &= \text{dist}(B_S s, \text{Ran } B_N)^2 \\ &= \|P B_S s\|^2 = \langle Q_S s, s \rangle. \end{aligned}$$

Since A_S is injective, R_S is positive definite. Taking the infimum over $s \neq 0$ and applying Rayleigh–Ritz proves (89). $\square$

The projection P is essential: directions invisible to A may still be visible to B , and can then cancel part of $B_S s$ without changing the denominator. Any lower certificate which ignores this nuisance minimization is not the exact coefficient-uniform constant.

6.2 Exact norm ratios and the triangle certificate

For $Az \neq 0$, define the pointwise missing ratio and physical ratio

$$q(z) := \frac{\|Mz\|^2}{\|Az\|^2}, \quad r(z) := \frac{\|(A + M)z\|^2}{\|Az\|^2}. \quad (90)$$

The corresponding uniform missing ratio is the extended-real number

$$q_{\text{unif}} := \sup_{Az \neq 0} q(z) \in [0, \infty]. \quad (91)$$

In finite dimension there is an exact dichotomy:

$$\boxed{q_{\text{unif}} < \infty \iff \text{Ker } A \subseteq \text{Ker } M.} \quad (92)$$

Indeed, if $n \in \text{Ker } A$ but $Mn \neq 0$, choose s with $As \neq 0$. Then $A(s + tn) = As$, whereas $\|M(s + tn)\| \rightarrow \infty$ as $|t| \rightarrow \infty$, so $q_{\text{unif}} = \infty$. Conversely, if M vanishes on $\text{Ker } A$, both numerator and denominator depend only on $s \in (\text{Ker } A)^\perp$; compactness of the corresponding normalized sphere gives a finite supremum. The exact shorted constant remains well defined even in the infinite- q_{unif} case.

Let

$$\varphi(t) := (1 - \sqrt{t})_+^2 \quad (0 \leq t < \infty), \quad \varphi(\infty) := 0. \quad (93)$$

This function is nonincreasing.

Proposition 6.2 (Pointwise and uniform triangle bounds). *For every z with $Az \neq 0$,*

$$\boxed{r(z) \geq \varphi(q(z))}. \quad (94)$$

Uniformly over all coefficient directions,

$$\boxed{\delta_{\text{sh}}(A, M) \geq \varphi(q_{\text{unif}})}. \quad (95)$$

Neither inequality is an identity in general.

Proof. The reverse triangle inequality gives

$$\|(A + M)z\| \geq (\|Az\| - \|Mz\|)_+.$$

Divide by $\|Az\|$, square, and use (90) to obtain (94). Since $q(z) \leq q_{\text{unif}}$ and φ is nonincreasing, taking the infimum over z proves (95). $\square$

Two elementary examples show why the shorted constant is strictly more informative.

Example 6.3 (Angle information is discarded). *Take $\mathcal{K} = \mathcal{H} = \mathbb{C}$, $Az = z$, and $Mz = iz$. Then*

$$q_{\text{unif}} = 1, \quad \varphi(q_{\text{unif}}) = 0, \quad \delta_{\text{sh}}(A, M) = |1 + i|^2 = 2.$$

The norm of Mz alone does not record its favorable angle with Az .

Example 6.4 (An infinite uniform q can coexist with positive shorted reassembly). *Take $\mathcal{K} = \mathcal{H} = \mathbb{C}^2$ and*

$$A(x, n) = (x, 0), \quad M(x, n) = (0, n).$$

Then $q_{\text{unif}} = \infty$, so the triangle certificate is zero, but

$$\frac{\|(A + M)(x, n)\|^2}{\|A(x, n)\|^2} = 1 + \frac{|n|^2}{|x|^2} \geq 1.$$

Thus $\delta_{\text{sh}}(A, M) = 1$. The shorting projection removes the orthogonal output generated by the A -invisible coordinate.

6.3 The restricted native constant

Let $\mathbf{D} \geq 0$ be the raw atomic diagonal on $\mathcal{K}$, and write

$$\mathcal{D}(z) := \langle \mathbf{D}z, z \rangle. \quad (96)$$

We assume

$$\text{Ker } \mathbf{D} \subseteq \text{Ker } M, \quad (97)$$

as holds when $\mathbf{D} = L^*L$ is formed from the literal active atomic lift and M is obtained by grouping those atoms. Put $S_D := \text{Ran } \mathbf{D} = (\text{Ker } \mathbf{D})^\perp$.

If $S_D \neq \{0\}$, define

$$\boxed{\kappa_M := \lambda_{\max} \left(\mathbf{D}^{\dagger/2} M^* M \mathbf{D}^{\dagger/2} \Big|_{S_D} \right)}. \quad (98)$$

If $S_D = \{0\}$, set $\kappa_M = 0$. In the Rayleigh formula below the supremum over the empty set is likewise defined to be zero. Rayleigh–Ritz gives the exact equivalent formulas

$$\kappa_M = \sup_{\mathcal{D}(z) > 0} \frac{\|Mz\|^2}{\mathcal{D}(z)}, \quad (99)$$

$$= \inf \{ \kappa \geq 0 : M^* M \preceq \kappa \mathbf{D} \}. \quad (100)$$

Thus κ_M is the least coefficient-uniform constant in

$$\|Mz\|^2 \leq \kappa_M \mathcal{D}(z). \quad (101)$$

Indeed, for $\mathcal{D}(z) > 0$, set $x = \mathbf{D}^{1/2}z \in S_D$. By (97), replacing z by $\mathbf{D}^{\dagger/2}x$ does not change Mz , and hence

$$\frac{\|Mz\|^2}{\mathcal{D}(z)} = \frac{\langle \mathbf{D}^{\dagger/2} M^* M \mathbf{D}^{\dagger/2} x, x \rangle}{\|x\|^2}.$$

Taking the maximum Rayleigh quotient proves (99). A quadratic-form bound for all z is equivalent to $M^* M \preceq \kappa_M \mathbf{D}$, which proves (100) and the asserted optimality.

To compare $\mathcal{D}$ with represented mass, put

$$\eta(z) := \frac{\mathcal{D}(z)}{\|Az\|^2} \quad (Az \neq 0), \quad \eta_{\text{unif}} := \sup_{Az \neq 0} \eta(z) \in [0, \infty]. \quad (102)$$

The exact pointwise restricted quotient is

$$\kappa_M(z) := \begin{cases} \|Mz\|^2 / \mathcal{D}(z), & \mathcal{D}(z) > 0, \\ 0, & \mathcal{D}(z) = 0. \end{cases} \quad (103)$$

The second line is consistent with (97). Therefore

$$q(z) = \kappa_M(z) \eta(z), \quad 0 \leq \kappa_M(z) \leq \kappa_M. \quad (104)$$

6.4 Residual and ambient upper bounds

There are two standard ways to upper-bound κ_M , and they need not be comparable with one another.

First suppose the missing map factors through a residual-labelled space,

$$M = \mathbf{E}T, \quad T : \mathcal{K} \rightarrow \mathcal{R}, \quad \mathbf{E} : \mathcal{R} \rightarrow \mathcal{H} \quad (105)$$

and assume $\text{Ker } \mathbf{D} \subseteq \text{Ker } T$, as for a residual grouping of the same literal atomic lift. Define

$$\kappa_{\text{res}} := \sup_{\mathcal{D}(z) > 0} \frac{\|Tz\|^2}{\mathcal{D}(z)}, \quad (106)$$

$$\beta_M := \sup_{0 \neq y \in \text{Ran } T} \frac{\|\mathbf{E}y\|^2}{\|y\|^2}. \quad (107)$$

Zero maps are assigned constant zero. Direct substitution gives

$$\boxed{\kappa_M \leq \kappa_{\text{res}} \beta_M.} \quad (108)$$

The restriction of β_M to the actual image $\text{Ran } T$ is essential; adjoining unused residual directions can only make the certificate coarser.

Second, suppose the literal atomic factorization is

$$\mathbf{D} = L^* L, \quad M = \mathbf{C}L, \quad (109)$$

where $\mathbf{C}$ sums the active atomic coordinates in each native output fiber. If $\mathcal{F}_i$ is the active fiber at native output i , set

$$d_{\text{missing}} := \max_i |\mathcal{F}_i|. \quad (110)$$

When there is no active fiber, this maximum is defined to be zero. Fiberwise Cauchy–Schwarz is sharp on the free atomic space and gives $\|\mathbf{C}\|^2 = d_{\text{missing}}$. Since $L\mathbf{D}^{\dagger/2}|_{S_D}$ is an isometry,

$$\boxed{\kappa_M \leq d_{\text{missing}}.} \quad (111)$$

Both (108) and (111) are upper certificates for the same optimal restricted constant; neither is a second loss to be added to it.

6.5 The dominance partial order

For lower certificates in the same scope, say that C_1 dominates C_2 when $C_1 \geq C_2$ for every admissible instance. Thus the larger number is the stronger guaranteed lower bound.

Theorem 6.5 (Uniform certificate lattice). *Assume $\eta_{\text{unif}} < \infty$. Then*

$$q_{\text{unif}} \leq \kappa_M \eta_{\text{unif}} \leq \begin{cases} \kappa_{\text{res}} \beta_M \eta_{\text{unif}}, \\ d_{\text{missing}} \eta_{\text{unif}}. \end{cases} \quad (112)$$

Consequently the corresponding coefficient-uniform lower certificates satisfy

$$\boxed{\delta_{\text{sh}}(A, M) \geq \varphi(q_{\text{unif}}) \geq \varphi(\kappa_M \eta_{\text{unif}}) \geq \begin{cases} \varphi(\kappa_{\text{res}} \beta_M \eta_{\text{unif}}), \\ \varphi(d_{\text{missing}} \eta_{\text{unif}}). \end{cases}} \quad (113)$$

The restricted certificate therefore dominates each coarse certificate. The residual-product and ambient-degree branches are generally incomparable; they form a fork, not a linear chain.

Proof. By (101), for every $Az \neq 0$,

$$q(z) = \frac{\|Mz\|^2}{\|Az\|^2} \leq \kappa_M \frac{\mathcal{D}(z)}{\|Az\|^2} \leq \kappa_M \eta_{\text{unif}}.$$

Taking the supremum proves the first inequality in (112). The remaining two follow from (108) and (111). Apply the nonincreasing function φ , and use proposition 6.2, to obtain (113). The explicit strict examples below prove incomparability of the two terminal branches. $\square$

The lattice records three distinct information losses. Passing from δ_{sh} to q_{unif} discards the angle and the exact nuisance elimination. Passing from q_{unif} to $\kappa_M \eta_{\text{unif}}$ separates two suprema which may be attained on different coefficient directions. Passing from κ_M to either coarse branch enlarges the actual coefficient image.

6.6 Explicit incomparability models

We now construct two literal finite grouping models. In both, the raw atomic norm is $\mathcal{D}(z) = \|Lz\|^2$, residual labels are retained by T , and E erases them. Hence the examples compare the actual constants in (108) and (111), rather than arbitrary numerical upper bounds.

Example 6.6 (The residual product can be strictly smaller than the ambient degree). *Let $\mathcal{K} = \mathbb{C}$, let the free atomic space be $\mathbb{C}^2$, and let two atoms share one native output but carry distinct residual labels. Define*

$$Lz = \frac{z}{\sqrt{2}}(1, i), \quad T = L, \quad E(y_1, y_2) = y_1 + y_2. \quad (114)$$

*Then $\mathbf{D} = L^*L = 1$, $\kappa_{\text{res}} = 1$, and $\text{Ran } T = \mathbb{C}(1, i)$. Therefore*

$$\beta_M = \frac{|1+i|^2}{2} = 1, \quad \kappa_{\text{res}} \beta_M = 1. \quad (115)$$

The free native grouping has a fiber of size two, so $d_{\text{missing}} = 2$. The physical map is $Mz = (1+i)z/\sqrt{2}$, and hence $\kappa_M = 1$. Thus

$$\kappa_M = \kappa_{\text{res}} \beta_M = 1 < 2 = d_{\text{missing}}. \quad (116)$$

Both atoms are active; the saving comes from restricting the erasure to the actual one-dimensional coefficient image, not from deleting an atom.

Example 6.7 (The ambient degree can be strictly smaller than the residual product). *Fix integers $r, k \geq 2$. Take a free atomic space with two native fibers. Fiber $i = 1$ contains r atoms, all with residual label $s = 1$. Fiber $i = 2$ contains k atoms with distinct residual labels $s = 1, \dots, k$. Let $\mathcal{K}$ be the full atomic space and $L = I$, so $\mathbf{D} = I$.*

Let T first sum atoms having the same pair (i, s) , and let E then sum residual labels at each fixed i . The first map has squared norm

$$\kappa_{\text{res}} = r, \quad (117)$$

attained in the size- r group at $(1, 1)$. The map T is onto its $(1 + k)$ -dimensional residual output: the first grouped coordinate and each of the k singleton coordinates can be prescribed independently. Consequently

$$\beta_M = k, \quad \kappa_{\text{res}}\beta_M = rk. \quad (118)$$

The composed map $M = ET$ is simply native grouping in the two fibers. Its squared norm and its ambient degree are

$$\kappa_M = d_{\text{missing}} = \max\{r, k\}. \quad (119)$$

Thus

$$d_{\text{missing}} = \max\{r, k\} < rk = \kappa_{\text{res}}\beta_M. \quad (120)$$

The two stage norms are attained in different native fibers, which is why multiplying them loses information.

Together, [examples 6.6](#) and [6.7](#) prove that no universal arrow exists between $\kappa_{\text{res}}\beta_M$ and d_{missing} . In contrast, the arrows from both of them down to the optimal restricted constant κ_M are universal and sharp.

6.7 Distinguished vectors and scope preservation

The preceding lattice is coefficient-uniform. For one distinguished physical vector z_* with $Az_* \neq 0$, the correct pointwise lattice is different.

Proposition 6.8 (Distinguished-vector certificate lattice). *Let z_* satisfy $Az_* \neq 0$. Then*

$$\begin{aligned} r(z_*) &\geq \varphi(q(z_*)) = \varphi(\kappa_M(z_*)\eta(z_*)) \\ &\geq \varphi(\kappa_M\eta(z_*)) \end{aligned} \quad (121)$$

$$\geq \begin{cases} \varphi(\kappa_{\text{res}}\beta_M\eta(z_*)), \\ \varphi(d_{\text{missing}}\eta(z_*)). \end{cases} \quad (122)$$

The number $\kappa_M(z_)$ is an exact Rayleigh quotient for this one vector. A favorable value of it cannot replace κ_M in [theorem 6.5](#).*

Proof. The first inequality is [\(94\)](#); the equality is [\(104\)](#). Since $\kappa_M(z_*) \leq \kappa_M \leq \kappa_{\text{res}}\beta_M$ and $\kappa_M \leq d_{\text{missing}}$, monotonicity of φ gives all remaining inequalities. A single Rayleigh quotient supplies no Loewner bound on other directions, so it cannot imply the uniform replacement. $\square$

This distinction can be arbitrarily large. Let $A = I$, $\mathbf{D} = I$, and $M = \text{diag}(0, L)$ on $\mathbb{C}^2$. At $z_* = e_1$, $\kappa_M(z_*) = 0$, while the uniform constant is $\kappa_M = L^2$. Hence a perfect distinguished certificate may coexist with an arbitrarily poor uniform one.

6.8 How to read the lattice

The exact shorted constant is the decision quantity for coefficient- uniform reassembly. The remaining nodes are sufficient lower certificates. Under the standing hypothesis $\eta_{\text{unif}} < \infty$ of [theorem 6.5](#), the partial order is

$$\delta_{\text{sh}}(A, M) \geq \varphi(q_{\text{unif}}) \geq \varphi(\kappa_M \eta_{\text{unif}}) \geq \begin{cases} \varphi(\kappa_{\text{res}} \beta_M \eta_{\text{unif}}), \\ \varphi(d_{\text{missing}} \eta_{\text{unif}}). \end{cases} \quad (123)$$

The two bottom nodes are not ordered. Improving either one is useful only as a way to bound κ_M ; once κ_M itself is known, neither coarser number is charged in addition. Likewise, a pointwise value $q(z_*)$ or $\kappa_M(z_*)$ belongs only to the distinguished chain, whereas q_{unif} , κ_M , and δ_{sh} belong to the uniform chain.

All conclusions here are L0 finite-dimensional identities and inequalities. They neither estimate the physical TPC constants nor show that any of them is subpower in X . In particular, the lattice does not establish cancellation in the affine Möbius kernel, a fixed-shift parity gain, a prime-pair lower square, or a twin-prime statement.

7 Localized alias inverse theorems

TPC-61 identifies the exact normalized missing Gram, but deliberately does not estimate its off-diagonal arithmetic kernel [\[11\]](#). This section records the converse information that is available without any new number theory: if the restricted native norm is large, then a large portion of that norm must already be visible on one row, on one literal collision cell, and finally on one row pair. Thus a failure of a proposed norm bound cannot remain completely delocalized.

Let F_M be the normalized physical missing incidence matrix on the active row space, and put

$$\mathbf{K}_M := F_M^* F_M, \quad \mathbf{H}_M := \mathbf{K}_M - I, \quad \kappa_M := \lambda_{\max}(\mathbf{K}_M). \quad (124)$$

Every column of F_M has norm one. Hence

$$(\mathbf{K}_M)_{m,m} = 1, \quad (\mathbf{H}_M)_{m,m} = 0, \quad \kappa_M = 1 + \lambda_{\max}(\mathbf{H}_M). \quad (125)$$

The rows in this section are only the actually active rows. No zero row is adjoined to inflate a degree or a dimension.

7.1 Literal collision-cell decompositions

We use a decomposition that is exact at the level of Gram summands. Let $\mathcal{B}$ be a nonempty set of B literal collision cells and write

$$\mathbf{H}_M = \sum_{b \in \mathcal{B}} \mathbf{H}_M^{(b)}. \quad (126)$$

Here every common-output product of two distinct source rows is assigned to exactly one cell, together with its conjugate product, and $(\mathbf{H}_M^{(b)})_{m,m} = 0$. Thus [\(126\)](#) is an identity with the literal phases; it is not a support majorant. If the original packet is first partitioned into source cells, then $\mathcal{B}$ must include all nonempty cell pairs that can contribute a Gram product. In particular, B below counts correlation pieces, not merely the number of source-cell names. The decomposition is fixed by the literal carrier before any coefficient vector or extremizing eigenvector is selected.

For a row m and a cell b , set

$$\mathcal{N}_b(m) := \{n \neq m : (\mathbf{H}_M^{(b)})_{m,n} \neq 0\}, \quad \Delta := \max_{m,b} |\mathcal{N}_b(m)|. \quad (127)$$

This is the degree of the actual cellwise collision graph. It is not an ambient number of labels that never occur.

Theorem 7.1 (Localized alias inverse theorem). *Assume the active row space is nonzero and*

$$\kappa_M \geq 1 + L \quad (128)$$

for some $L > 0$. Then all of the following hold.

(i) *There is an active row m such that*

$$\sum_{n \neq m} |(\mathbf{K}_M)_{m,n}| \geq L. \quad (129)$$

(ii) *There are an active row m and a literal cell $b \in \mathcal{B}$ such that*

$$\sum_{n \neq m} |(\mathbf{H}_M^{(b)})_{m,n}| \geq \frac{L}{B}. \quad (130)$$

(iii) *If $\Delta < \infty$, then for some $m \neq n$ and some $b \in \mathcal{B}$,*

$$|(\mathbf{H}_M^{(b)})_{m,n}| \geq \frac{L}{B\Delta}, \quad (131)$$

$$\left(\sum_{n \in \mathcal{N}_b(m)} |(\mathbf{H}_M^{(b)})_{m,n}|^2 \right)^{1/2} \geq \frac{L}{B\sqrt{\Delta}}. \quad (132)$$

When $L > 0$, the conclusion itself forces $\Delta \geq 1$.

Proof. The Hermitian matrix $\mathbf{H}_M$ has zero diagonal. Its largest eigenvalue is at most its spectral radius, and the spectral radius is at most the induced ℓ^∞ -norm. Therefore

$$L \leq \lambda_{\max}(\mathbf{H}_M) \leq \max_m \sum_{n \neq m} |(\mathbf{H}_M)_{m,n}|.$$

This proves (129), because the off-diagonal entries of $\mathbf{H}_M$ and $\mathbf{K}_M$ agree.

Choose a row m satisfying (129). The exact cell decomposition and the triangle inequality give

$$L \leq \sum_{n \neq m} \left| \sum_{b \in \mathcal{B}} (\mathbf{H}_M^{(b)})_{m,n} \right| \leq \sum_{b \in \mathcal{B}} \sum_{n \neq m} |(\mathbf{H}_M^{(b)})_{m,n}|.$$

At least one cell has the mass in (130). That cell has at most Δ nonzero neighbours, so its largest entry is at least its ℓ^1 -mass divided by Δ , proving (131). Finally Cauchy–Schwarz on the same star gives

$$\frac{L}{B} \leq \sum_{n \in \mathcal{N}_b(m)} |(\mathbf{H}_M^{(b)})_{m,n}| \leq \sqrt{\Delta} \left(\sum_{n \in \mathcal{N}_b(m)} |(\mathbf{H}_M^{(b)})_{m,n}|^2 \right)^{1/2},$$

which is (132). □

The theorem is one-sided. A large cell entry is compatible with cancellation between cells and therefore need not force a comparably large entry of the completed Gram. What the theorem says is that a large *completed* norm forces at least one large literal cell sum.

7.2 Degree times pair correlation

The forward Schur estimate is the exact counterpart of the inverse theorem.

Corollary 7.2 (Pair correlation plus active degree). *Suppose, uniformly over all colliding distinct active rows, that*

$$|\{n \neq m : (\mathbf{K}_M)_{m,n} \neq 0\}| \leq X^{\delta+o(1)}, \quad (133)$$

$$|(\mathbf{K}_M)_{m,n}| \leq X^{-\tau+o(1)}, \quad (134)$$

where the two $o(1)$ terms are uniform in the row indices. Then

$$\boxed{\kappa_M \leq 1 + X^{\delta-\tau+o(1)}, \quad \nu_M \leq (\delta - \tau)_+,} \quad (135)$$

where $\nu_M := \limsup_{X \rightarrow \infty} \log \max\{1, \kappa_M\} / \log X$. In particular, a pair saving with $\tau < \delta$ does not by itself give a subpower operator bound: this Schur certificate still permits the positive exponent $\delta - \tau$. It does not assert that the actual ν_M is positive.

Proof. Every off-diagonal absolute row sum is at most $X^{\delta-\tau+o(1)}$. The same induced-norm estimate used in [theorem 7.1](#) yields the first assertion. If $\delta \leq \tau$, the right side is $X^{o(1)}$; otherwise its exponent is $\delta - \tau$. This proves the second assertion. $\square$

If instead the available estimate is cellwise, with $B = X^{b+o(1)}$, cell degree $X^{\delta+o(1)}$, and $|(\mathbf{H}_M^{(b)})_{m,n}| \leq X^{-\tau+o(1)}$, then the same proof gives

$$\nu_M \leq (b + \delta - \tau)_+. \quad (136)$$

The factor B may be omitted only when orthogonality or another exact summation mechanism removes it.

Conversely, suppose

$$\kappa_M \geq 1 + X^{\nu-o(1)}, \quad B \leq X^{b+o(1)}, \quad \Delta \leq X^{\delta+o(1)}. \quad (137)$$

Then [theorem 7.1](#) produces one cell and one row pair for which

$$|(\mathbf{H}_M^{(b)})_{m,n}| \geq X^{\nu-b-\delta-o(1)}, \quad (138)$$

$$\left(\sum_{n \in \mathcal{N}_b(m)} |(\mathbf{H}_M^{(b)})_{m,n}|^2 \right)^{1/2} \geq X^{\nu-b-\delta/2-o(1)}. \quad (139)$$

These are deterministic localization statements; no probability model is being invoked.

7.3 The arithmetic content of the localized witness

Conjugating $\mathbf{H}_M$ and every cell piece by the same fixed diagonal source-divisor gauge preserves the decomposition, every entry magnitude, and all row/star norms above. After this TPC-61 gauge, each direct-slice contribution to an off-diagonal entry is exactly the normalized, fixed- h_0 two-affine Möbius sum

$$\frac{1}{\sqrt{W_M(m)W_M(m')}} \sum_j \mathbf{m}_m(j) \mathbf{m}_{m'}(j) \mu(mj + h_0) \mu(m'j + h_0) \overline{a_m(j)} a_{m'}(j) \Gamma_{m,m'}(j). \quad (140)$$

The full off-diagonal entry is the sum of the left and right direct-slice expressions on their declared orthogonal output components; no cross-slice Gram term is inserted. Equivalently, the same sum is indexed by the literal solutions of

$$m'pr - mp'r' = h_0(m' - m), \quad (141)$$

with all inherited largest-prime, cofactor, missing-carrier, orbit, and endpoint restrictions retained. Restricting the Gram summands to a literal collision cell gives the entries $(\mathbf{H}_M^{(b)})_{m,m'}$ above. Consequently (138) localizes a large restricted native norm to one normalized signed determinant sum. It does not localize to one determinant quadruple, because many quadruples may still occur in that sum.

This is an exact reformulation, not a cancellation theorem. TPC-61 does not prove (134) for any $\tau > 0$, either for one pair uniformly or for the growing matrix of row pairs. The present inverse theorem likewise supplies no sign cancellation: it only says where a large norm must leave a witness. The matrix statements are L0, their literal fixed-shift identification is L1, and no L2 parity or prime-pair improvement is claimed.

8 A three-module conditional closure theorem

The master constant in section 3 is the exact answer for a complete finite-scale map. A proof rarely estimates it in one step. This section gives the precise modular statement that permits three lower constants to be multiplied. Its hypotheses include the carrier and quantifier compatibility that is often suppressed in an informal ledger.

8.1 Compatible form chains

Let $Z_X \subseteq \mathcal{H}_X$ be either a declared coefficient class or the singleton $\{z_*(X)\}$. Let

$$\mathcal{Q}_0, \quad \mathcal{Q}_A, \quad \mathcal{Q}_F, \quad \mathcal{Q}_{\text{out}} \tag{142}$$

be nonnegative quadratic forms on the same original coefficient space. Their intended meanings are parent input, represented output, full represented-plus-missing output, and downstream reconnected output.

Definition 8.1 (Compatible three-module chain). The four forms in (142) form a compatible chain on Z_X if:

- (i) every form is computed from the same initial vector z ;
- (ii) the output carrier and raw normalization used in one comparison are literally the input carrier and normalization used in the next;
- (iii) all maps and supports are fixed before z , for a uniform chain;
- (iv) every adaptive label is retained in the next output, or else a separate proved reassembly inequality is inserted;
- (v) no inequality is imported from a different shift, an averaged family, or an independently chosen coefficient vector.

Theorem 8.2 (Three-module closure). *Suppose the chain is compatible on Z_X . Let $c_A(X)$, $\delta_{\text{reasm}}(X)$, and $c_{\text{out}}(X)$ be finite nonnegative constants, and suppose*

$$\mathcal{Q}_A(z) \geq c_A(X) \mathcal{Q}_0(z), \tag{143}$$

$$\mathcal{Q}_F(z) \geq \delta_{\text{reasm}}(X) \mathcal{Q}_A(z), \tag{144}$$

$$\mathcal{Q}_{\text{out}}(z) \geq c_{\text{out}}(X) \mathcal{Q}_F(z) \tag{145}$$

for every $z \in Z_X$. Then

$$\boxed{\mathcal{Q}_{\text{out}}(z) \geq c_{\text{out}}(X) \delta_{\text{reasm}}(X) c_A(X) \mathcal{Q}_0(z)} \tag{146}$$

for every $z \in Z_X$.

If Z_X is the full declared coefficient class, the conclusion is uniform. If $Z_X = \{z_*(X)\}$, it is distinguished only.

Proof. If $\mathcal{Q}_0(z) = 0$, or if any of the three constants is zero, the right-hand side of (146) is zero and the conclusion follows from nonnegativity of $\mathcal{Q}_{\text{out}}$. Otherwise all three constants are positive. The first lower bound then makes $\mathcal{Q}_A(z) > 0$, and the second makes $\mathcal{Q}_F(z) > 0$. The three inequalities may therefore be applied successively to the actual intermediate forms generated by the same z . Multiplication gives (146). No quantifier changes in this substitution. $\square$

The compatibility conditions are necessary for this proof. For example, three statements of the form “there exists a good vector” need not concern the same vector. Likewise, a phase sector chosen after the first arrow need not be the fixed input map assumed by the second.

8.2 Exponent form and endpoint condition

Suppose, in one compatible scope, that $\Lambda_A, \lambda_{\text{reasm}}, \Lambda_{\text{out}} \geq 0$, and that all $o(1)$ terms below are uniform over that declared scope:

$$c_A(X) \geq X^{-\Lambda_A + o(1)}, \quad (147)$$

$$\delta_{\text{reasm}}(X) \geq X^{-\lambda_{\text{reasm}} + o(1)}, \quad (148)$$

$$c_{\text{out}}(X) \geq X^{-\Lambda_{\text{out}} + o(1)}. \quad (149)$$

Then

$$\mathcal{Q}_{\text{out}}(z) \geq X^{-(\Lambda_A + \lambda_{\text{reasm}} + \Lambda_{\text{out}}) + o(1)} \mathcal{Q}_0(z). \quad (150)$$

The inherited endpoint comparison has room only under

$$\boxed{\Lambda_A + \lambda_{\text{reasm}} + \Lambda_{\text{out}} < \frac{1}{400}.} \quad (151)$$

This is a conditional closure of the declared truncated fixed- h_0 packet lower square. It does not identify $\mathcal{Q}_{\text{out}}$ with a prime-pair counting function, prove a boundary/tail comparison, or establish any of the three exponent inputs.

8.3 Admissible reassembly inputs

The middle constant in (146) may be obtained in several honest ways.

Master shorted-Gram input. Let R_A represent $\mathcal{Q}_A$, and let $F = A_X + M_X$ represent $\mathcal{Q}_F$. If $R_A \neq 0$, then the optimal full-space uniform value is the master constant

$$\delta_{\text{sh}} = \alpha(F; R_A) \quad (152)$$

from theorem 3.2. A quantitative lower bound for δ_{sh} may be inserted directly as δ_{reasm} . The kernel inclusion proves only positivity at each finite scale, not a polynomial exponent. Here “optimal” refers to the full coefficient space, or to a homogeneous declared class containing a nonzero scalar multiple of every coefficient direction. On an arbitrary restricted class the same full-space constant remains a valid lower bound, but need not be optimal for that class. If $R_A = 0$, then $\mathcal{Q}_A$ is identically zero, the middle comparison has no positive denominator, and the reassembly arrow is vacuous; the master constant is not invoked or assigned a value.

Restricted mass-degree input. On the same literal carrier, let $\sigma_M, \lambda_{\text{rep}} \geq 0$, suppose the effective growth exponent ν_M defined in (11) is finite, and assume

$$\mathcal{Q}_A(z) = \|A_X z\|^2, \quad \mathcal{Q}_F(z) = \|(A_X + M_X)z\|^2, \quad \|M_X z\|^2 \leq \kappa_M \mathcal{D}_M(z), \quad (153)$$

together with

$$\mathcal{D}_M(z) \leq X^{-\sigma_M+o(1)} \mathcal{D}_{\text{ref}}(z), \quad (154)$$

$$\kappa_M \leq X^{\nu_M+o(1)}, \quad (155)$$

$$\mathcal{Q}_A(z) \geq X^{-\lambda_{\text{rep}}+o(1)} \mathcal{D}_{\text{ref}}(z) \quad (156)$$

in the same scope. For vectors with $\mathcal{Q}_A(z) > 0$, define

$$q_X(z) := \frac{\kappa_M \mathcal{D}_M(z)}{\mathcal{Q}_A(z)} \leq X^{\nu_M+\lambda_{\text{rep}}-\sigma_M+o(1)}. \quad (157)$$

The strict condition

$$\boxed{\sigma_M > \nu_M + \lambda_{\text{rep}}} \quad (158)$$

therefore makes $q_X(z) = o(1)$. The reverse triangle inequality applied to (153) shows that one may take the lower certificate $\delta_{\text{reasm}}(X) = 1 - o(1)$, and hence $\lambda_{\text{reasm}} = 0$. Vectors with $\mathcal{Q}_A(z) = 0$ impose no condition on this middle lower bound. This conclusion is uniform only if all three inputs and their $o(1)$ terms are uniform; replacing κ_M by one Rayleigh quotient gives only a distinguished statement.

At equality of fixed-power exponents, let $a \geq 0$. If, uniformly over the same vectors with $\mathcal{Q}_A(z) > 0$,

$$1 - \sqrt{q_X(z)} \geq X^{-a+o(1)}, \quad (159)$$

then one may take $\delta_{\text{reasm}}(X) \geq X^{-2a+o(1)}$, so $\lambda_{\text{reasm}} \leq 2a$. The endpoint requires

$$\boxed{\Lambda_A + 2a + \Lambda_{\text{out}} < \frac{1}{400}}. \quad (160)$$

Residual-factor input. On the critical cofactor block, TPC-61 leaves the conditional algebraic window

$$e + \mu_{\text{erase}} + \lambda_{\text{rep}} < \frac{39671}{210000}. \quad (161)$$

If the inherited residual certificate supplies the corresponding same-scope missing-to-represented estimate on every one of the $O(\log X)$ blocks, if the global comparison loses at most this logarithmic factor, and if (161) holds with one fixed positive margin uniformly over the blocks, then one may take $\lambda_{\text{reasm}} = 0$. The logarithmic factor is subpower. The algebraic window by itself, or mere blockwise strictness with a margin tending to zero, is not enough.

The numerical right side here is much larger than $1/400$. Passing this test proves only that the middle reassembly exponent is zero. It does not place λ_{rep} , Λ_A , or the downstream losses inside the final endpoint budget; that independent check is still (151).

Signed distinguished input. For one physical vector, an exact lower estimate of

$$2 \operatorname{Re} \langle A_X z_*, M_X z_* \rangle + \|M_X z_*\|^2 \quad (162)$$

may prove a favorable distinguished reassembly ratio even when every uniform certificate is silent. It may not be entered as δ_{reasm} for the full coefficient class.

8.4 No double charging

The exponents σ_M , ν_M , μ_{erase} , e , and λ_{rep} in the preceding tests decide the single middle exponent λ_{reasm} . They are not five additional arrows. Moreover, if λ_{rep} is assembled from represented-chain losses already contained in Λ_A , it is used in (158) but is not added to Λ_A a second time. The same rule applies to every diagnostic capacity exponent.

Exact coordinate changes, a fixed large support constant in $y = C_{\text{vh}} Q$, Moore–Penrose restriction to an actual active support, and finite or logarithmic diagnostic partitions have zero fixed-power cost.

9 Exact independence models and scope counterexamples

The closure gates in this paper are logically distinct. The following finite-dimensional models show that raw missing mass, prime-block isometries, a strong represented map, and successful reassembly cannot be interchanged. Every coefficient and every atomic norm below is literal. The models are L0 interface examples; they are not asserted to arise from the fixed- h_0 arithmetic carrier and therefore are not L1 or L2 counterexamples.

9.1 One output and N distinct prime blocks

Fix $N \geq 2$. Let the row coefficient space be $\mathbb{C}^N$, let the native output space be $\mathbb{C}$, and define

$$A_N \zeta := \frac{1}{N} \sum_{m=1}^N \zeta_m. \quad (163)$$

For each row m , introduce one missing atom w_m . All N atoms land at the same native output, their terminal-prime labels $p_1, \dots, p_N$ are pairwise distinct, and their literal coefficients are

$$c_{w_m} = -\frac{1}{N}. \quad (164)$$

Thus the missing map and its raw atomic diagonal are

$$M_N \zeta = -\frac{1}{N} \sum_{m=1}^N \zeta_m = -A_N \zeta, \quad \mathbf{D}_M = \frac{1}{N^2} I_N. \quad (165)$$

Proposition 9.1 (Exact coherent missing model). *The model in (165) satisfies the fixed- (i, p) singleton property and the exact prime-block identities*

$$M_{p_m}^* M_{p_m} = \mathbf{D}_{p_m}, \quad \sum_{m=1}^N \mathbf{D}_{p_m} = \mathbf{D}_M. \quad (166)$$

Nevertheless

$$F_M = M_N \mathbf{D}_M^{-1/2} = [-1 \ \cdots \ -1], \quad \mathbf{K}_M = \mathbf{1}\mathbf{1}^*, \quad \kappa_M = N. \quad (167)$$

For the distinguished coefficient $\zeta_\star = \mathbf{1}$,

$$\|A_N \zeta_\star\|^2 = 1, \quad \langle \mathbf{D}_M \zeta_\star, \zeta_\star \rangle = \frac{1}{N}, \quad (A_N + M_N) \zeta_\star = 0. \quad (168)$$

Thus a raw missing fraction $1/N$ can coexist with complete reassembly failure.

Proof. For a fixed terminal prime p_m , the block map is

$$M_{p_m} \zeta = -\frac{1}{N} \zeta_m.$$

There is exactly one atom at the pair consisting of the unique output and p_m . Hence

$$M_{p_m}^* M_{p_m} = \frac{1}{N^2} e_m e_m^* = \mathbf{D}_{p_m}.$$

Summing over m proves (166). Since $\mathbf{D}_M^{-1/2} = N I_N$, the normalized incidence matrix has N unit columns, all equal to -1 . Its Gram is $\mathbf{1}\mathbf{1}^*$, whose eigenvalues are $N, 0, \dots, 0$. This proves (167). The three identities in (168) follow by direct substitution. $\square$

The mass-degree product is exactly critical in this example:

$$\frac{\kappa_M \langle \mathbf{D}_M \zeta_\star, \zeta_\star \rangle}{\|A_N \zeta_\star\|^2} = 1. \quad (169)$$

The reverse-triangle margin is therefore zero, and the actual operators attain that worst case. The example does not say that equality must cause cancellation; it shows why a theorem based only on this certificate needs a strict margin or additional phase information.

9.2 A uniformly perfect represented map can still cancel

The rank-one nature of A_N in the preceding example is not the source of the obstruction. Let both the coefficient and output spaces be $\mathbb{C}^N$, put

$$v := N^{-1/2}\mathbf{1}, \quad \tilde{A}_N := I_N, \quad \tilde{M}_N := -vv^* = -\frac{1}{N}\mathbf{1}\mathbf{1}^*. \quad (170)$$

Realize $\tilde{M}_N$ by one atom $w_{i,m}$ of coefficient $-1/N$ for every output-row pair (i, m) . Give the N^2 atoms pairwise distinct formal terminal-prime labels. Then every fixed- (i, p) incidence is a singleton, each one-atom prime block is an exact isometry relative to its atomic diagonal, and

$$\tilde{\mathbf{D}}_M = \frac{1}{N}I_N, \quad \tilde{F}_M = -\frac{1}{\sqrt{N}}\mathbf{1}\mathbf{1}^*, \quad \tilde{\mathbf{K}}_M = \mathbf{1}\mathbf{1}^*, \quad \tilde{\kappa}_M = N. \quad (171)$$

The represented map is uniformly optimal:

$$\|\tilde{A}_N\zeta\|^2 = \|\zeta\|^2 \quad (\zeta \in \mathbb{C}^N). \quad (172)$$

Yet on the unit vector v ,

$$\langle \tilde{\mathbf{D}}_M v, v \rangle = \frac{1}{N}, \quad (\tilde{A}_N + \tilde{M}_N)v = 0. \quad (173)$$

Thus even a coefficient-uniform represented lower bound, literal raw missing mass, and all singleton prime-block identities do not imply a positive reassembled lower bound. Control of the restricted native Gram or of the signed cross term is an independent gate.

The same model also separates uniform and distinguished conclusions. For every $\zeta \perp v$, one has

$$\tilde{M}_N\zeta = 0, \quad (\tilde{A}_N + \tilde{M}_N)\zeta = \zeta. \quad (174)$$

Hence the large uniform value $\tilde{\kappa}_M = N$ does not prove failure for a specified physical vector. Conversely, good reassembly on one such vector gives no uniform lower bound because the direction v is still dark. Quantifier scope cannot be changed in either direction.

9.3 Pair decorrelation and collision degree are independent

For $0 \leq \rho \leq 1$, take orthonormal vectors $e_0, e_1, \dots, e_N$ and define the unit columns

$$f_m := \sqrt{1-\rho}e_m + \sqrt{\rho}e_0 \quad (1 \leq m \leq N). \quad (175)$$

Their Gram is

$$\mathbf{K}_\rho = (1-\rho)I_N + \rho\mathbf{1}\mathbf{1}^*, \quad \lambda_{\max}(\mathbf{K}_\rho) = 1 + (N-1)\rho. \quad (176)$$

Every pair correlation is only ρ , but every row has $N-1$ neighbours. If $N = X^{\delta+o(1)}$ and $\rho = X^{-\tau+o(1)}$, this family has exact Gram exponent

$$\limsup_{X \rightarrow \infty} \frac{\log \lambda_{\max}(\mathbf{K}_\rho)}{\log X} = (\delta - \tau)_+. \quad (177)$$

It therefore shows that the degree factor in [corollary 7.2](#) cannot be removed at the Gram level. Small pair correlations and sparse collision support are separate inputs.

9.4 What the models do and do not refute

The examples establish exact logical independence at the finite-dimensional operator interface:

- (i) prime-block isometry does not control unequal-prime cross terms;
- (ii) small raw missing mass does not control native missing energy without the restricted factor κ_M ;
- (iii) a uniform represented lower bound does not prevent phase-aligned cancellation by the missing map;
- (iv) a large uniform κ_M does not force failure on a distinguished coefficient vector; and
- (v) pairwise decorrelation does not imply an operator bound without an active collision-degree estimate.

They do not violate the exact TPC-61 formulas. Rather, they attain those formulas in abstract incidence systems. They also do not satisfy or refute the full fixed- h_0 two-affine determinant restrictions. Consequently they prove no negative theorem about the literal arithmetic packet and no parity, prime-pair, or twin-prime conclusion. Their role is narrower and quantifier-safe: any successful physical closure theorem must use a hypothesis that excludes the displayed coherent configurations in the same scope as its conclusion.

10 Endpoint arithmetic, kill tests, and pivot rules

The abstract theorems identify which constants are exact and which certificates dominate others. This section places the three numerical comparisons inherited from TPC-60–TPC-61 into that decision structure. They are route tests, not new arithmetic estimates.

10.1 Three numerical comparisons

At the critical cofactor scale,

$$\theta_J = \frac{133}{400}, \quad \lambda_L = \frac{99979}{210000}. \quad (178)$$

The coefficient-blind residual determinant-ladder exponent is

$$\lambda_L - \theta_J = \frac{15077}{105000}. \quad (179)$$

Combined with the balanced cofactor saving, it leaves the conditional residual window

$$\boxed{2\theta_J - \lambda_L = \frac{39671}{210000}}. \quad (180)$$

Thus the residual mass-degree certificate can be perturbative if

$$e + \mu_{\text{erase}} + \lambda_{\text{rep}} < \frac{39671}{210000} \quad (181)$$

with one uniform fixed margin over the compatible dyadic blocks. Since $39671/210000 \gg 1/400$, this inequality is only a test for zero-cost reassembly. A value of λ_{rep} may satisfy this test and still make the represented-chain loss Λ_A too large for the final endpoint. It is charged in the final ledger exactly when it is part of a represented-chain arrow, and never as a second reassembly arrow.

The direct ambient native-degree route has instead the positive gap

$$\boxed{\frac{267}{400} - \frac{133}{400} = \frac{67}{200}}. \quad (182)$$

It cannot make the missing perturbation $o(1)$ by combining the full ambient degree with the strongest nonempty pure-cardinality saving. By the certificate order in [section 6](#), this closes only that coarse ambient proof. It neither lower-bounds the restricted κ_M nor implies failure of the residual, signed, or shorted routes.

Finally, the localized-alias theorem gives a different conditional door. Let $\delta, \tau \geq 0$. If the actual normalized collision graph has row degree

$$\Delta_X \leq X^{\delta+o(1)} \quad (183)$$

and every off-diagonal literal normalized pair entry satisfies, uniformly for colliding $m \neq n$,

$$|K_{m,n}| \leq X^{-\tau+o(1)}, \quad (184)$$

then

$$\nu_M \leq (\delta - \tau)_+. \quad (185)$$

The optimal restricted mass-degree certificate consequently has the strict sufficient condition

$$\boxed{\sigma_M > \lambda_{\text{rep}} + (\delta - \tau)_+}. \quad (186)$$

provided the degree, pair, missing-mass, and represented-reference estimates all hold on the same literal carrier and in the same quantifier scope. No favorable value of δ , τ , or σ_M is proved in this paper.

10.2 The complete physical ledger

Expanding the represented and downstream modules gives the inherited loss sum. Every quantity denoted by λ below is a nonnegative loss exponent:

$$\boxed{\begin{aligned} \Lambda_{\text{tot}} := & \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} \\ & + \lambda_H + \lambda_{\text{aff}} + \lambda_{\text{reasm}} + \lambda_{\text{bdry}} + \lambda_{\text{tail}}. \end{aligned}} \quad (187)$$

The endpoint condition is

$$\boxed{\Lambda_{\text{tot}} < \frac{1}{400}}. \quad (188)$$

In the three-module notation,

$$\Lambda_A = \lambda_{\text{sel}} + \lambda_{\text{occ}} + \lambda_{\text{blk}} + \lambda_{\text{term}} + \lambda_{\text{sing}} + \lambda_H + \lambda_{\text{aff}}, \quad \Lambda_{\text{out}} = \lambda_{\text{bdry}} + \lambda_{\text{tail}}. \quad (189)$$

Every displayed identity, exact gauge, canonical relabeling, and diagnostic finite or logarithmic partition costs zero. The quantities in (181) and (186) determine whether λ_{reasm} is zero or positive; they are not added separately to (187).

For a distinguished chain, the reassembly loss is replaced consistently by $\lambda_{\text{reasm},*}$, and every upstream and downstream inequality must concern the same physical vector. Equality in (188) is rejected unless leading constants and all remainders recover a strict final inequality.

10.3 Exact kill and pivot table

The word “kill” is reserved for a proved zero in the object actually needed by the route. The following table separates exact deaths from certificate failures.

Observed event	What is killed	Honest pivot
$\text{Ker } F_X \not\subseteq \text{Ker } R_X$	The full-coefficient-space master transfer; its exact constant is zero.	Change scope to a declared distinguished vector, or change the actual physical theorem.
A mandatory factor T_X has a kernel vector of positive parent energy	Every final map that genuinely factors through T_X .	None within the same mandatory factorization; reopen the physical map.
An active row has $W_R(m) = 0 < W_M(m)$, and its row direction lies in the declared uniform class	Uniform represented raw-mass transfer on that class.	Reopen the retained-relation provenance, or use a distinguished vector known to avoid that row.
$\text{Ker}(A_X + M_X) \not\subseteq \text{Ker } A_X$	Full-coefficient-space uniform represented-to-full reassembly.	A distinguished signed estimate may remain possible.
The sharp singleton floor is zero	The singleton sufficient certificate only.	Estimate the complete coherent fiber or use another literal certificate.
$\eta_{\text{frozen}} = 0 < \eta_{\text{adaptive}}$	One frozen selector as a uniform map.	Retain all selector labels by direct sum, or prove frozen-selector stability and then reassembly.
$\sigma_M < \nu_M + \lambda_{\text{rep}}$, or equality holds without a proved leading-constant margin	The zero-fixed-power conclusion from the exponent-only restricted mass-degree test.	Test the exact shorted Gram, a signed distinguished vector, or improve the exposure/alias inputs.
The ambient gap is 67/200	The full ambient-degree phase-blind proof.	Use the actual restricted Gram, residual quotient, or signed shorting.
A necessary proved loss reaches 1/400	The stated endpoint exponent route without a constant-level recovery.	Change the upstream architecture; do not hide the loss in an $X^{o(1)}$ term.

10.4 Operational order for the next arithmetic attack

The preceding tests should be applied in the following order.

- Door A.** *Master kernels.* Check the kernel inclusions for every mandatory map before estimating small eigenvalues.
- Door B.** *Literal exposure.* Recover the actual retained relation and compute $W_R(m)$, $W_M(m)$, and their physical distinguished weighting.
- Door C.** *Represented chain.* Prove critical occupancy, terminal activation, and represented native transfer in one declared scope.
- Door D.** *Alias localization.* Compute actual row/cell degrees and either prove a pair/star estimate for the TPC-61 determinant kernel or exhibit a large obstruction witness.
- Door E.** *Reassembly decision.* Apply the optimal restricted product test, the residual alternative, or the exact shorted/signed route.

Door F. *Downstream reconnection.* Prove boundary and tail arrows and then audit the strict complete ledger.

This order prevents a difficult parity estimate from being attempted on an empty represented carrier and prevents an adaptive or distinguished success from being promoted to a uniform map.

10.5 Forbidden repairs

None of the killed routes is repaired by replacing h_0 , averaging over shifts, dividing raw atoms by fiber sizes, deleting coefficient-dependent zeros, choosing atom phases, or substituting one favorable Rayleigh quotient for a uniform generalized eigenvalue. Those operations define different objects. They may be interesting in their own right, but their conclusions cannot be entered into (187) for the fixed carrier.

11 Result ledger, claim boundary, and the next theorem

TPC-62 turns the TPC-53-TPC-61 interfaces into an exact decision system. It does not fill the remaining arithmetic inputs. This final section separates the two achievements.

11.1 Result ledger

Object	Exact conclusion	Status
Master transfer	The optimal full-coefficient-space uniform constant from a nonzero semidefinite parent form to a final map is the projected shorted-Gram eigenvalue in (4).	L0 theorem
Kernel criterion	Positive finite-scale full-space uniform transfer holds exactly when $\text{Ker } F \subseteq \text{Ker } R$.	L0 iff test
Nested nuisance elimination	Shorting to nested retained spaces is transitive; sequential exact elimination equals one master shorting.	L0 theorem
Mandatory/certificate split	A mandatory factor with a positive-energy kernel direction kills every final map factoring through it; a zero sufficient-certificate constant need not kill the physical map.	L0 decision
Selector quantifiers	Frozen selector max–min is at most adaptive min–max, with a sharp coordinate-projection gap.	L0 theorem
Labeled uniformization	Retaining all selector labels by direct sum has exact constant $\lambda_{\min}(\sum_a S_a^* S_a)$; label erasure is a new map.	L0 theorem
Certificate order	The exact shorted constant dominates valid reverse-triangle bounds; restricted κ_M dominates residual and ambient relaxations, which are generally incomparable.	L0 partial order
Alias localization	$\kappa_M \geq 1 + L$ forces a row of alias mass L , a literal cell of mass L/B , and a pair/star witness after a degree bound.	L0 inverse theorem
Degree–pair bridge	Row degree $X^{\delta+o(1)}$ and pair bound $X^{-\tau+o(1)}$ imply $\nu_M \leq (\delta - \tau)_+$.	L0 theorem; L1 target
Literal alias entry	Each TPC–61 entry is the fixed- h_0 weighted two-affine Möbius correlation, equivalently its determinant-cell sum.	L1 identity
Three-module closure	Compatible represented, reassembly, and downstream lower constants multiply; exponents must sum strictly below $1/400$.	L0 theorem; fixed- h_0 application conditional L1
Independence models	Small raw missing mass, prime-block isometry, or good upstream constants do not separately prevent exact coherent reassembly cancellation.	L0 sharpness

These statements reduce ambiguity. They do not make an unproved input positive.

11.2 What is not proved

The following conclusions are absent.

1. The master arithmetic constant α_X is not shown to be bounded below by $X^{-\lambda+o(1)}$ for any $\lambda < 1/400$.
2. The actual retained relation is not shown to have positive exposure on every active row, nor is the missing mass shown to save a fixed power in a uniform or distinguished scope.
3. Critical-window occupancy, terminal activation, singleton survival, represented native transfer, boundary reconnection, and the tail estimate are not jointly established on one carrier.

4. No frozen phase sector or dyadic block is shown to work uniformly for every coefficient vector. Adaptive selection is not relabeled uniform.
5. The actual normalized collision graph is not shown to have degree $X^{\delta+o(1)}$ with a favorable δ .
6. No new upper bound is proved for the weighted correlation $\mu(mj + h_0)\mu(m'j + h_0)$, for its determinant form, or for a dyadic row-star.
7. Consequently $\kappa_M = X^{o(1)}$, $\nu_M < \sigma_M - \lambda_{\text{rep}}$, and $\lambda_{\text{reasm}} = 0$ are not proved for the physical family.
8. No positive fixed-shift parity estimate, prime-pair lower bound, prime-pair asymptotic, or twin-prime implication is obtained.

In particular, h_0 has not been specialized to 2. Merely making that substitution in an L1 identity would not supply the missing carrier, correlation, reassembly, downstream, or target-identification theorems.

In particular, the phrase “localized alias obstruction” means that a large eigenvalue must leave a literal witness. It does not say that the witness is small. Likewise, the conditional closure theorem concludes a lower bound only for the declared truncated packet form named in its hypotheses. Identifying that form with a number-theoretic target requires the separately listed downstream theorem.

11.3 The next arithmetic theorem

The new calculus narrows the next useful theorem to a concrete alternative.

Route 1. *Exposure first.* Reopen the construction of the retained relation $\mathcal{I}_X$ and prove, in one declared scope,

$$\mathcal{D}_M(z) \leq X^{-\sigma_M+o(1)}\mathcal{D}_{\text{ref}}(z), \quad \sigma_M > 0. \quad (190)$$

Uniformly this requires every active row; distinguishedly it requires the exact weights $W_H(m)|z_m|^2$.

Route 2. *Represented reference in the same scope.* Establish

$$\|A_X z\|^2 \geq X^{-\lambda_{\text{rep}}+o(1)}\mathcal{D}_{\text{ref}}(z) \quad (191)$$

after proving the actual critical occupancy and terminal activation used to define A_X .

Route 3. *Alias degree and cancellation.* On literal dyadic determinant cells, prove either a pair/star estimate

$$|K_{m,n}| \leq X^{-\tau+o(1)} \quad (m \neq n), \quad \Delta_X \leq X^{\delta+o(1)}, \quad (192)$$

strong enough that

$$\sigma_M > \lambda_{\text{rep}} + (\delta - \tau)_+, \quad (193)$$

or compute a large row/cell witness that closes this subroute and directs attention to exact shorting.

Route 4. *Exact reassembly alternative.* If the degree-pair route is too weak, estimate the master shorted Gram or the signed physical cross term directly. Apply the kernel tests before any asymptotic eigenvalue analysis.

Route 5. *Downstream theorem.* Only after one reassembly route is quantified should the boundary and tail maps be reconnected and the full strict ledger (187) be tested.

The first two routes prevent a sophisticated alias estimate from being spent on an unrepresented carrier. The third turns every large operator obstruction into an explicit determinant cell, so failure is informative rather than merely numerical.

11.4 Possible outcomes

The next layer has three publishable outcomes with different strengths.

- (i) A favorable exposure and alias estimate opens the restricted mass-degree route and supplies a zero-cost reassembly input. It gives a conditional L1 packet lower bound only after the compatible represented and downstream modules also pass the strict ledger.
- (ii) A rank or kernel witness closes its matching architecture exactly; a localized-cell witness that violates the proposed norm hypothesis closes that narrower subroute and supplies an explicit pivot.
- (iii) A theorem valid only for adaptive selectors, averaged shifts, or a distinguished vector remains useful in that scope, provided it is not promoted to the uniform fixed-shift route.

Conclusion. The TPC-53–TPC-61 sequence has not crossed the parity barrier. It has replaced a loosely connected heuristic with a literal packet, exact finite-dimensional gates, sharp no-go tests, and one explicit remaining alias kernel. TPC-62 adds the master closure constant, quantifier calculus, certificate order, and localization inverse theorem. The surviving road is therefore narrower and falsifiable: prove actual cofactor exposure, prove a represented reference, and then control or exhibit one literal alias row/cell strongly enough to pass the single reassembly gate. That is a precise next problem, not a claim that it has already been solved.

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